

»» Chapter V ««

CAUCHY THEORY

PREFATORY REMARKS

A prime fact of the Cauchy theory may be stated in terms already clear to us:
An analytic function admits locally a primitive or, put more explicitly, if f is analytic at a , then $f| \Delta(a; r)$ admits a primitive, r small and positive.

We shall see that the converse is also true (cf. §7, this chapter). The Cauchy–Goursat theorem for a triangle, which will occupy our attention in §2, this chapter, in fact yields the local primitive property of an analytic function (and is readily concluded from it). The essential steps in basing the Cauchy theory consist in first establishing a local principle, such as the Cauchy–Goursat theorem for a triangle, and then passing from it to a global principle. How far one goes in an initial study is a matter of taste since the great classical developments and applications of the theory unfold beautifully with very little apparatus. In §3, this chapter, we follow the approach of F. M. Stewart to show that we may obtain a global principle starting from the local primitive property. The way is paved for further developments. For the homology-theoretic approach, which we shall not develop, we refer to the initial account of Ahlfors [1] and the treatment of Dinghas [38]. The homotopy-theoretic approach, to be given here, has been used by H. Cartan [26] and Dieudonné [36].

Once the general theory has been developed, we find that the integral theorems obtained have numerous applications. At the earliest stages we have tools that permit us to evaluate certain integrals (such as those of Fresnel) and to treat such matters as Ankeny’s proof of the Fundamental Theorem of

Algebra (Ex. 3.29, this chapter) and the theorem of Riesz-Fejér in its simplest form (Ex. 3.30, this chapter). Most importantly, we are led to integral representation formulas for analytic functions. The advance is great, for we may conclude the striking results that the analytic functions coincide with those that are (locally) of power series type and that a function f analytic on an open set is the uniform limit *in* its domain of a sequence of rational functions, each analytic in the domain of f . Further, such formulas lead to the results that the uniform limit of a sequence of analytic functions is analytic and that a nonconstant function analytic on a region is an *open* map, that is, carries open sets into open sets. Later we shall see that the Cauchy theory has a wide range of applicability.

The achievement of Goursat in the Cauchy theory was to show that the *mere* stipulation of the existence of a derivative of a function at each point of its domain, an open set in $\mathbf{C}$, implied the validity of the Cauchy theorem for a triangle for the function. Under a C' hypothesis the question becomes much simpler. Indeed, suppose that f is analytic on a square

$$I = \{|\operatorname{Re}(z - \alpha)| < h, |\operatorname{Im}(z - \alpha)| < h\},$$

h positive, and f' is continuous. Let F be the associate of f . Then (F, iF) satisfies the exactness condition of §6, Ch. IV, by virtue of the Cauchy-Riemann condition. Let G satisfy $G_x = F$, $G_y = iF$. Then G is the associate of g analytic on I and $g' = f$. We conclude that the C' hypothesis implies the local primitive property.

Our first step will be to treat the concept of a *line integral*. As preliminaries we shall want the notion of rectifiability and the Riemann-Stieltjes integral. While it may be argued that the generality of the notion of rectifiability is not needed for the elementary Cauchy theory, it enters very naturally in the existence theorem for the line integral.

1. LINE INTEGRALS

We start with a study of the simplest facts concerning Riemann-Stieltjes integrals.

Variation. This notion depends on some method of marking off stages on a given segment of $\mathbf{R}$ which we shall take as not reducing to a singleton, observing the distance between the images of successive stages with respect to a given map whose domain is the segment, and thereupon summing the so obtained distances. It is in the nature of things to take the image space as metric. There is no reason at the outset to restrict attention to maps into $\mathbf{R}$ or $\mathbf{C}$. Let $a, b \in \mathbf{R}$, $a < b$. We term a finite subset P of $[a, b]$ a *partitioning* of

$[a, b]$ provided that $a, b \in P$. There exists a unique increasing finite sequence $(x_k)_0^m$ with domain an initial section of $\mathbf{N}$ whose image is P . It is termed the *sequence associated with P* . We have: $x_0 = a, x_m = b$. By the *norm* of P , denoted $|P|$, is meant $\max\{x_{k+1} - x_k : k = 0, \dots, m-1\}$. Let $f: [a, b] \rightarrow X$ where X is a metric space with metric ρ . By the *P -variation of f* , denoted $v(f; P)$, is meant

$$\sum_0^{m-1} \rho[f(x_k), f(x_{k+1})].$$

By the (*total*) *variation of f* , denoted $V[f]$, is meant the supremum of the set of P -variations of f , all P being taken into account. A partitioning Q of $[a, b]$ is termed a *refinement of P* provided that $Q \supset P$. We shall have frequent occasion in this section to compare sums associated with a partitioning and corresponding sums associated with a refinement. To that end, the following device is useful. Let $(y_k)_0^n$ denote the unique increasing finite sequence with domain an initial section of $\mathbf{N}$ whose image is Q , a refinement of P . Let $v(k)$ satisfy $y_{v(k)} = x_k, k = 0, \dots, m$, so that $v(0) = 0, v(m) = n$, and $v(k) < v(k+1)$. We have

$$v(f; Q) = \sum_0^{m-1} \left\{ \sum_{v(k)}^{v(k+1)-1} \rho[f(y_j), f(y_{j+1})] \right\}. \quad (1.1)$$

By the triangle inequality we conclude that

$$v(f; P) \leq v(f; Q).$$

Length. To follow traditional language, we shall use rather than “(total) variation” the term *length* for a path $\gamma: [a, b] \rightarrow X$ and shall write $L[\gamma]$ for $V[\gamma]$. A path γ is termed *rectifiable* provided that $L[\gamma] < +\infty$.

EXERCISES

1.1. If $a < c < b$ and $f_1 = f|_{[a, c]}$ and $f_2 = f|_{[c, b]}$, then $V[f] = V[f_1] + V[f_2]$.

1.2. Suppose that $X = \mathbf{C}$ and ρ is the standard metric, $\rho(z, w) = |w - z|$. Show that with $f, g: [a, b] \rightarrow \mathbf{C}, c \in \mathbf{C}$, we have:

- (a) $V[\operatorname{Re} f], V[\operatorname{Im} f] \leq V[f] \leq V[\operatorname{Re} f] + V[\operatorname{Im} f]$.
- (b) $V[f + g] \leq V[f] + V[g]$.
- (c) $V[cf] = |c| V[f]$.
- (d) $V[f g] \leq (\sup_{[a, b]} |f|) V[g] + (\sup_{[a, b]} |g|) V[f]$.

In (c) and (d) the convention $(+\infty)(0) = (0)(+\infty) = 0$ is to hold.

- (e) If $|f(x)| \geq m > 0, a \leq x \leq b$, then $V[1/f] \leq m^{-2} V[f]$.

*(f) If f' is continuous on $[a, b]$, then

$$V[f] \leq (\max_{[a, b]} |f'|)(b - a).$$

Hint: For (d) the identity $BD - AC = (B - A)D + A(D - C)$ is useful.

Riemann-Stieltjes Integral. We proceed with a, b as above and $f, g: [a, b] \rightarrow \mathbb{C}$ and form sums given in terms of f, g , a partitioning P of $[a, b]$ whose associated sequence is $(x_k)_0^m$, and a finite sequence $u = (u_k)_0^{m-1}$ satisfying $x_k \leq u_k \leq x_{k+1}$, $k = 0, \dots, m-1$. Such a u will be termed *P-admitted*. We introduce

$$S(P; u) = \sum_0^{m-1} f(u_k) [g(x_{k+1}) - g(x_k)]$$

and consider the behavior of $S(P; u)$ for $|P|$ small. We say that f possesses a *Riemann-Stieltjes integral with respect to g* provided that there exists $A \in \mathbb{C}$ such that for each positive η there exists a positive δ such that

$$|S(P; u) - A| < \eta$$

when $|P| < \delta$. Of course, reference is made to *P-admitted* u . It is immediate that there is at most one such A . In case such A exists, we term A the *Riemann-Stieltjes integral of f with respect to g* and denote it

$$\int_a^b f dg \quad (\text{or} \quad \int_a^b f(x) dg(x)).$$

When $g(x) = x$, $a \leq x \leq b$, we term the integral simply the *Riemann integral* of f , denoting it

$$\int_a^b f dx \quad (\text{or} \quad \int_a^b f(x) dx).$$

In general, given f, g as above and that $a \leq \alpha \leq \beta \leq b$, by

$$\int_\alpha^\beta f dg$$

we understand 0 when $\alpha = \beta$ and the integral of $f|[\alpha, \beta]$ with respect to $g|[\alpha, \beta]$ when $\alpha < \beta$, assuming that it exists.

Since we are concerned with developing the notion of the Riemann-Stieltjes integral for our specific needs in complex function theory, we turn directly to the important classical case where f is continuous on $[a, b]$ and $V[g] < +\infty$ and show

Theorem 1.1: *If f is continuous on $[a, b]$ and $V[g] < +\infty$, then f possesses a Riemann–Stieltjes integral with respect to g and*

$$\left| \int_a^b f dg \right| \leq (\max_{[a,b]} |f|) V[g].$$

PROOF: We introduce ω , the modulus of continuity of f , and proceed to compare the $S(P; u)$. Suppose that Q is a refinement of P and that t is Q -admitted. Following the cue given by (1.1) and using the $v(k)$ notation, we see that

$$S(Q; t) - S(P; u) = \sum_{j=0}^{m-1} \sum_{v(k)}^{v(k+1)-1} [f(t_j) - f(u_k)][g(y_{j+1}) - g(y_j)],$$

where $(y_k)_0^{v(m)}$ is the sequence associated with Q . Hence

$$|S(Q; t) - S(P; u)| \leq \omega(|P|) V[g].$$

Let P and Q now be arbitrary partitionings of $[a, b]$. Using $R = P \cup Q$ and z that is R -admitted we see that

$$\begin{aligned} |S(Q; t) - S(P; u)| &\leq |S(Q; t) - S(R; z)| + |S(R; z) - S(P; u)| \\ &\leq [\omega(|Q|) + \omega(|P|)] V[g]. \end{aligned} \quad (1.2)$$

Now consider

$$P_n = \{a + 2^{-n}k(b-a); k = 0, \dots, 2^n\}, \quad n \in \mathbb{N},$$

and

$$u_n = (a + 2^{-n}k(b-a))_0^{2^n-1}.$$

Let $S_n = S(P_n; u_n)$. We see that

$$|S_n - S_{n+k}| \leq \omega[2^{-n}(b-a)] V[g], \quad k \in \mathbb{N}.$$

Hence (S_n) is a Cauchy sequence. Let $A = \lim_{n \rightarrow \infty} S_n$. Using (1.2), we see that for arbitrary P and P -admitted u we have

$$\begin{aligned} |S(P; u) - A| &\leq |S(P; u) - S_n| + |S_n - A| \\ &\leq \{\omega(|P|) + \omega[2^{-n}(b-a)]\} V[g] + |S_n - A|, \end{aligned}$$

whence, taking the limit as $n \rightarrow \infty$, we have

$$|S(P; u) - A| \leq \omega(|P|) V[g]. \quad (1.3)$$

The inequality (1.3) shows that f possesses an integral with respect to g and brings to light very vividly the roles played by the continuity of f and the

finiteness of $V[g]$. The inequality of the theorem follows at once from (1.3). The theorem is established.

EXERCISES

1.3. In this exercise f (with or without subscript) denotes a complex-valued function with domain $[a, b]$, continuous on $[a, b]$; g (with or without subscript) denotes a complex-valued function with domain $[a, b]$, satisfying $V[g] < +\infty$; $k \in \mathbb{C}$. Show the following. (a) If $a < c < b$, then

$$\int_a^b f dg = \int_a^c f dg + \int_c^b f dg.$$

(b) We have

$$\begin{aligned}\int_a^b (f_1 + f_2) dg &= \int_a^b f_1 dg + \int_a^b f_2 dg; \\ \int_a^b f d(g_1 + g_2) &= \int_a^b f dg_1 + \int_a^b f dg_2; \\ \int_a^b (kf) dg &= \int_a^b f d(kg) = k \int_a^b f dg.\end{aligned}$$

[Bilinearity.] (c) If (f_n) tends to f uniformly on $[a, b]$, then

$$\lim_{n \rightarrow \infty} \int_a^b f_n dg = \int_a^b f dg.$$

(d) Let $G(a) = 0$, $G(x) = V[g | [a, x]]$, $a < x \leq b$. Then G is never-decreasing on $[a, b]$ (that is, $G(x) \leq G(y)$ when $x < y$; “never-increasing” similarly) and

$$\left| \int_a^b f dg \right| \leq \int_a^b |f| dG.$$

When $g(x) = x$, $a \leq x \leq b$, then

$$\int_a^b |f| dG = \int_a^b |f| dx.$$

(e) Suppose that φ is a never-decreasing continuous map of $[A, B]$ onto $[a, b]$, $A, B \in \mathbb{R}$. Then

$$\int_A^B f \circ \varphi d(g \circ \varphi) = \int_a^b f dg.$$

(f) Given a partitioning P of $[a, b]$ with associated sequence $(x_k)_0^m$ and u that is P -admitted. Then

$$\left| \sum_{k=0}^{m-1} g(x_k) [f(x_{k+1}) - f(x_k)] + \int_a^b f dg - [f(b)g(b) - f(a)g(a)] \right| \leq 2\omega(|P|) V[g];$$

$$\left| \sum_{k=0}^{m-1} [g(u_k) - g(x_k)][f(x_{k+1}) - f(x_k)] \right| \leq \omega(|P|) V[g].$$

Hint:

$$\begin{aligned} & g(x_k) [f(x_{k+1}) - f(x_k)] + f(x_k) [g(x_{k+1}) - g(x_k)] \\ &= [f(x_{k+1})g(x_{k+1}) - f(x_k)g(x_k)] - [f(x_{k+1}) - f(x_k)][g(x_{k+1}) - g(x_k)]. \end{aligned}$$

Consequently,

$$\int_a^b g df$$

exists and

$$\int_a^b g df + \int_a^b f dg = f(b)g(b) - f(a)g(a)$$

(integration by parts). If f' is continuous on $[a, b]$, then

$$\int_a^b gf' dx$$

exists and is equal to

$$\int_a^b g df.$$

This last assertion holds also when g is continuous on $[a, b]$. Hint: In both cases compare $S(P; u)$ for (gf', x) and for (g, f) .

1.4. With f continuous on $[a, b]$, the function F given by

$$F(x) = \int_a^x f(t) dt, \quad a \leq x \leq b,$$

is a primitive of f . Hence if G is a primitive of f ,

$$\int_a^b f(x) dx = G(b) - G(a).$$

1.5. Review the earlier uses of the Riemann integral in this book. Formulate explicitly the properties used and justify them in the framework of the present treatment of the integral.

1.6. Let X be a compact metric space and let f be a continuous map of $X \times [a, b]$ into $\mathbb{C}$. Let g be as initially given in Ex. 1.3. Show that the function

$$\int_a^b f(x, t) dg(t)$$

with domain X is uniformly approximated by sums of the form

$$\sum_{k=0}^{m-1} f(x, t_k) [g(t_{k+1}) - g(t_k)]$$

for $|P|$ small; that is, show that given ε positive, there exists a positive δ such that when $|P| < \delta$, the difference of sum and integral has absolute value $< \varepsilon$, $x \in X$. Conclude that the integral defines a continuous function on X (cf. Ex. 5.10, Ch. III). If $X = \Delta(c; r)$, $0 < r < +\infty$, and if

$$\lim_c \frac{f(z, t) - f(c, t)}{z - c} = F(t), \quad a \leq t \leq b,$$

uniformly on $[a, b]$ (that is, given positive ε , there exists a positive δ such that the difference of $F(t)$ and the difference quotient on the left is in modulus less than ε when $0 < |z - c| < \delta$ for all $t \in [a, b]$), then

$$\int_a^b F(t) dg(t)$$

is the derivative at c of

$$\int_a^b f(z, t) dg(t).$$

The variation of g when g' is continuous on $[a, b]$. We shall establish under the stated assumption the useful formula

$$V[g] = \int_a^b |g'| dx.$$

PROOF: From

$$g(\beta) - g(\alpha) = \int_\alpha^\beta g' dx, \quad a \leq \alpha \leq \beta \leq b,$$

using (d) and (a) of Ex. 1.3, this section, we see that

$$v(g; P) \leq \int_a^b |g'| dx,$$

and hence

$$V[g] \leq \int_a^b |g'| dx.$$

On the other hand, with $(x_k)_0^m$ denoting the sequence associated with P we have

$$v(g; P) - \sum_0^{m-1} |g'(x_k)| (x_{k+1} - x_k) = \sum_0^{m-1} \left[\left| \int_{x_k}^{x_{k+1}} g' dx \right| - \left| \int_{x_k}^{x_{k+1}} g'(x_k) dx \right| \right].$$

Hence, using the standard inequalities for the absolute values of sums and differences, we have

$$\left| v(g; P) - \sum_0^{m-1} |g'(x_k)| (x_{k+1} - x_k) \right| \leq \sum_0^{m-1} \left| \int_{x_k}^{x_{k+1}} [g'(x) - g'(x_k)] dx \right|.$$

Consequently,

$$\sum_0^{m-1} |g'(x_k)| (x_{k+1} - x_k) \leq v(g; P) + \omega(|P|) (b - a),$$

where ω now denotes the modulus of continuity of g' . We conclude

$$\int_a^b |g'| dx \leq V[g].$$

The asserted equality follows.

EXERCISES

1.7. $\gamma(t) = \text{cis } t$ on $[0, a]$, $a > 0$. We have $L[\gamma] = a$.

1.8. Let $\gamma(0) = 0$, $\gamma(t) = t + it^\alpha \sin(\pi/t)$, $0 < t \leq 1$. Show that γ is rectifiable when $\alpha > 1$, but is not when $0 < \alpha \leq 1$.

Line integrals. Let γ be a rectifiable path in $\mathbb{C}$ with domain $[a, b]$ and let $f: E \rightarrow \mathbb{C}$, where E contains the image of γ . Suppose that $f \circ \gamma$ is a continuous map of $[a, b]$ into $\mathbb{C}$. By the *line integral of f along γ* , denoted

$$\int_{\gamma} f dz,$$

is meant

$$\int_a^b (f \circ \gamma) d\gamma.$$

Let $f: A \rightarrow \mathbb{C}$ where A is a region in $\mathbb{C}$ and f is continuous on A . Let γ be a rectifiable path in A with domain $[a, b]$. Suppose that g is a primitive of f . Then the following very simple formula holds:

$$\int_{\gamma} f dz = g[\gamma(b)] - g[\gamma(a)]. \quad (1.4)$$

If γ' is continuous on $[a, b]$, we have

$$\int_{\gamma} f dz = \int_a^b (g \circ \gamma)' dt$$

and hence (1.4).

We assume, as we may, that the domain of γ is $[0, 1]$. It suffices to replace γ by $\gamma[a + (b - a)t]$, $0 \leq t \leq 1$. Suppose that A is of the form $\Delta(c; r)$, so that $|\gamma(t) - c| \leq \rho < r$, $0 \leq t \leq 1$ for some ρ , $0 < \rho < r$. Let P be a partitioning of $[0, 1]$, $(t_k)_0^m$ its associated sequence, and Γ the path satisfying

$$\begin{aligned} \Gamma(t) &= \gamma(t_k) \frac{t_{k+1} - t}{t_{k+1} - t_k} + \gamma(t_{k+1}) \frac{t - t_k}{t_{k+1} - t_k}, \\ t_k &\leq t \leq t_{k+1}, \quad k = 0, \dots, m-1. \end{aligned}$$

Now we conclude using (1.4), which is available for Γ restricted to each $[t_k, t_{k+1}]$, that

$$\int_{\Gamma} f dz = g[\gamma(1)] - g[\gamma(0)].$$

Noting that

$$\int_{\Gamma} f dz - \sum_0^{m-1} f[\gamma(t_k)][\gamma(t_{k+1}) - \gamma(t_k)] = \sum_0^{m-1} \int_{t_k}^{t_{k+1}} [f \circ \Gamma - (f \circ \Gamma)(t_k)] d\Gamma,$$

we conclude that

$$\left| g[\gamma(1)] - g[\gamma(0)] - \sum_0^{m-1} f[\gamma(t_k)][\gamma(t_{k+1}) - \gamma(t_k)] \right| \leq \omega_F[\omega_{\gamma}(|P|)] L[\gamma], \quad (1.5)$$

where $F = f|_{\overline{\Delta(c; \rho)}}$ and ω with subscript is the modulus of continuity of the function denoted by the subscript. We see that (1.4) follows from (1.5) under the assumption that A is a disk. To establish the result unrestrictedly we consider $A \neq \mathbf{C}$, the case of equality having been treated, and let μ denote

$$\min\{|\gamma(t) - w| : 0 \leq t \leq 1, w \in \mathbf{C} - A\} > 0$$

(cf. Ex. 5.8, Ch. III). Let $n \in \mathbf{N}$ be so large that $\omega_{\gamma}(2^{-n}) < \mu$. Then

$$|\gamma(t) - \gamma(t_k)| < \mu, \quad t_k \leq t \leq t_{k+1}, \quad k = 0, \dots, 2^n - 1,$$

where $t_k = k/2^n$. Consequently, the integral of f along $\gamma|_{[t_k, t_{k+1}]}$ is $g[\gamma(t_{k+1})] - g[\gamma(t_k)]$. Summing we obtain (1.4).

EXERCISES

1.9. Given γ a rectifiable path in $\mathbb{C}$, $n \in \mathbb{Z}$, $n \neq -1$, $a \in \mathbb{C}$, $a \notin \gamma$ when $n < 0$. Calculate $\int_{\gamma} (z - a)^n dz$.

1.10. Given γ a rectifiable path in $\mathbb{C}$, $c \in \mathbb{C} - \gamma$, $T(z) = Az + B$ where $A \neq 0$. Show that

$$\int_{\gamma} (z - c)^{-1} dz = \int_{T \circ \gamma} [z - T(c)]^{-1} dz.$$

1.11. Let $0 < a < b < +\infty$. Let $\gamma(t) = a \cos t + ib \sin t$, $0 \leq t \leq 2\pi$. Calculate $\int_{\gamma} z^{-1} dz$. Hint: Use the complex exponential expressions for $\cos$ and $\sin$.

2. THE SIMPLICIAL CAUCHY-GOURSAT THEOREM

We now treat the theorem usually called “the Cauchy–Goursat theorem for a triangle.” From the point of view of complex function theory it is of no great importance to know that certain of the sets we encounter are in non-degenerate cases triangles in the sense of 2-dimensional Euclidean geometry. The whole question can be treated directly without appeal to prior developments concerning plane Euclidean geometry. The range of applicability of the present simplicial version of the Cauchy–Goursat theorem will become clear from the exercises at the end of the section. In particular, the identity of analytic functions and functions of power series type will follow (Ex. 2.8, this section). The C'' assumption of Theorem 8.1 of Ch. IV is assured for analytic functions.

Given an ordered triple of complex numbers $T = (a, b, c)$, we let $\gamma_1 = \overrightarrow{ab}$, $\gamma_2 = \overrightarrow{bc}$, $\gamma_3 = \overrightarrow{ca}$ and introduce $l(T) = \sum_1^3 L[\gamma_k]$ and $d(T)$, the diameter of the convex hull of $\{a, b, c\}$.† Actually,

$$l(T) = |b - a| + |c - b| + |a - c|,$$

as is readily verified, and

$$d(T) = \max\{|b - a|, |c - b|, |a - c|\}.$$

To see the latter equality, let z be a point of the convex hull of $\{a, b, c\}$ so that $z = \lambda a + \mu b + \nu c$, where $\lambda, \mu, \nu \geq 0$ and $\lambda + \mu + \nu = 1$, and let w also be a point of the convex hull of $\{a, b, c\}$. Now

$$w - z = \lambda(w - a) + \mu(w - b) + \nu(w - c).$$

† Given $Y \subset X$ a metric space with metric ρ , by the *diameter* of Y is meant

$$\sup \{\rho(y_1, y_2) : y_1, y_2 \in Y\}.$$

Hence $|w - z| \leq \max\{|w - a|, |w - b|, |w - c|\}$. Applying this inequality with w replacing z and a (respectively, b, c) replacing w , we conclude that

$$|w - z| \leq \max\{|b - a|, |c - b|, |a - c|\},$$

and thereupon the formula for $d(T)$. We define the *quadrisections* T_1, T_2, T_3, T_4 of T by

$$\begin{aligned} T_1 &= (a, (a+b)/2, (c+a)/2), & T_2 &= ((a+b)/2, b, (b+c)/2), \\ T_3 &= ((b+c)/2, c, (c+a)/2), & T_4 &= ((c+a)/2, (a+b)/2, (b+c)/2). \end{aligned}$$

The adjoined figure indicates the state of affairs.

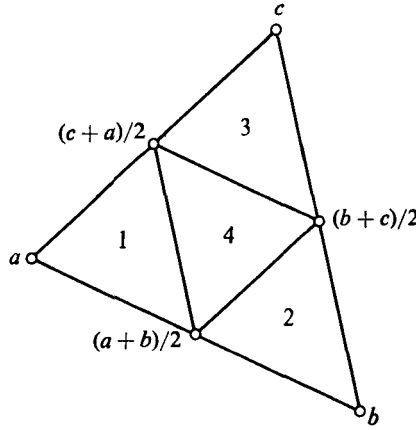


FIG. 5.1

Let O be an open subset of $\mathbb{C}$. T is termed *O -admissible* provided that the convex hull of $\{a, b, c\}$ is contained in O . If T is O -admissible, then so are its quadrisections.

Given $f: O \rightarrow \mathbb{C}$, f continuous on O , T O -admissible, we define $I(T; f)$ to be

$$\sum_{k=1}^3 \int_{\gamma_k} f dz.$$

By direct verification we see that

$$I(T; f) = \sum_{k=1}^4 I(T_k; f).$$

Hence if j is the least k for which $|I(T_k; f)|$ is the maximum of

$$\{|I(T_1; f)|, \dots, |I(T_4; f)|\},$$

we have

$$|I(T; f)| \leq 4|I(T_j; f)|.$$

The simplicial Cauchy–Goursat theorem is

Theorem 2.1: *If f is analytic on O , then $I(T; f) = 0$.*

PROOF: The O -admissibility of T plays an essential role. By a recursive argument we see that there exists a unique sequence $(T^{(n)})_{n \in \mathbf{N}}$ satisfying: $T^{(0)} = T$, $T^{(n+1)}$ is the quadrisection $T_j^{(n)}$ where j is taken as above, $T^{(n)}$ replacing T . It is easy to see that

$$l(T^{(n)}) = 2^{-n} l(T), \quad d(T^{(n)}) = 2^{-n} d(T),$$

and

$$|I(T; f)| \leq 4^n |I(T^{(n)}; f)|.$$

There exists a unique point w common to the convex hulls of the $T^{(n)}$, as follows when we note the nesting property of $(T^{(n)})_{n \in \mathbf{N}}$ namely, that the convex hull of $T^{(n+1)}$ is contained in the convex hull of $T^{(n)}$, $n \in \mathbf{N}$, and the fact that $\lim_{n \rightarrow \infty} d(T^{(n)}) = 0$. In fact, if z also satisfies this condition,

$$|w - z| \leq d(T^{(n)}), \quad n \in \mathbf{N};$$

whence $w = z$. To show existence we consider (w_n) , $w_n \in T^{(n)}$, and note that it is a Cauchy sequence. Its limit w has the asserted property. Now

$$f(z) = f(w) + f'(w)(z - w) + \varepsilon(z)(z - w),$$

where ε is continuous on O and $\varepsilon(w) = 0$. Hence

$$I(T^{(n)}; f) = I[T^{(n)}; \varepsilon(z)(z - w)],$$

since $f(w) + f'(w)(z - w)$ has a primitive. We conclude that

$$|I(T; f)| \leq 4^n M_n d(T^{(n)}) l(T^{(n)}) = M_n d(T) l(T),$$

where M_n is the maximum value taken by $|\varepsilon|$ on the convex hull of $T^{(n)}$. The theorem follows from the fact that $\lim_{n \rightarrow \infty} M_n = 0$.

EXERCISES

2.1. Establish the stated properties of $l(T^{(n)})$ and $d(T^{(n)})$.

2.2. Given $a, b, c, d \in \mathbf{R}$, $a \leq b$ and $c \leq d$. We denote by $[a, b; c, d]$ the set

$$\{[a(1-s) + bs] + i[c(1-t) + dt] : 0 \leq s, t \leq 1\}.$$

Suppose that $[a, b; c, d] \subset O$. Show by Theorem 2.1 that if f is analytic on O , then $J(b) = 0$, where

$$J(b) = \sum_1^4 \int_{\gamma_k} f dz,$$

$$\begin{aligned} \gamma_1 &= \overrightarrow{(a + ci)(b + ci)}, & \gamma_2 &= \overrightarrow{(b + ci)(b + di)}, \\ \gamma_3 &= \overrightarrow{(b + di)(a + di)}, & \gamma_4 &= \overrightarrow{(a + di)(a + ci)} \end{aligned}$$

(Cauchy–Goursat theorem for rectangle). Establish the asserted equality when f' is continuous on O by showing that $J'(b) = 0$.

Existence of primitive for the case of a disk. Suppose that f is analytic on $\Delta(a; r)$. Theorem 2.1 yields the existence of a primitive of f very rapidly. It is apparent that the convexity of $\Delta(a; r)$ is to our advantage. We define $g(z)$, $z \in \Delta(a; r)$, as the line integral of f along $\vec{az}$. Given $b \in \Delta(a; r)$, by Theorem 2.1 we have

$$g(z) - g(b) = \int_{\vec{bz}} f(w) dw,$$

noting that

$$g(z) = - \int_{\vec{za}} f(w) dw.$$

It follows that

$$g(z) - g(b) = f(b)(z - b) + \int_{\vec{bz}} [f(w) - f(b)] dw.$$

We conclude using the continuity of f at b and the inequality of Theorem 1.1, this chapter, that $g'(b) = f(b)$.

We state the result formally.

Theorem 2.2: *A function analytic on a disk possesses a primitive.*

EXERCISES

2.3. Show that if f is a complex-valued function whose domain is the convex hull of $\{a, b, c\} \subset \mathbb{C}$, where $\lim_z [f(w) - f(z)]/(w - z)$ exists at each point z of the domain of f , then $I(T; f) = 0$ where $T = (a, b, c)$ and the I notation is understood to be taken in the present context.

2.4. Given $f: O \rightarrow \mathbb{C}$ where O is an open subset of $\mathbb{C}$ and f is continuous on O . Suppose that f has the local primitive property: for each $z \in O$ there exists $r > 0$ for

which $\Delta(z; r) \subset O$ such that $f|_{\Delta(z; r)}$ possesses a primitive. Show that the conclusion of the simplicial Cauchy–Goursat theorem holds. Hint: Make sufficiently many repeated quadrisections.

2.5. Let $b \in O$ where O is an open subset of $\mathbb{C}$. Suppose that f is analytic on $O - \{b\}$ and that, in addition, f' is continuous on $O - \{b\}$. Suppose that

$$b \in \Delta(a; r) \subset \overline{\Delta(a; r)} \subset O, \quad 0 < r < +\infty.$$

Let

$$\begin{aligned} \varphi(s, t) &= b(1-t) + (a + re^{is})t, \\ 0 \leq s \leq 2\pi, \quad 0 < t \leq 1. \end{aligned}$$

For $t \in]0, 1[$ let $\Gamma_t(s) = \varphi(s, t)$, $0 \leq s \leq 2\pi$. Define $I(t)$ by

$$I(t) = \int_{\Gamma_t} f dz.$$

Conclude that I is constant by showing that $I'(t) = 0$, $0 < t \leq 1$.

2.6. Apply Ex. 2.5 to $[f(z) - f(b)]/(z - b)$ where f is analytic and f' is continuous on O , and conclude that

$$f(b) = \frac{1}{2\pi i} \int_{\Gamma_1} \frac{f(w)}{w - b} dw \quad (\text{Cauchy formula for disk}). \quad (2.1)$$

It is to be noted that

$$\int_{\Gamma_1} (w - b)^{-1} dw = 2\pi i, \quad (2.2)$$

as we see on using (5.3), Ch. IV. The formula (2.2) may also be concluded by using

$$\begin{aligned} (w - b)^{-1} &= [(w - a) - (b - a)]^{-1} \\ &= (w - a)^{-1} \left(1 - \frac{b - a}{w - a} \right)^{-1}, \quad w \in C(a; r), \end{aligned}$$

and the geometric series, which yields

$$(w - b)^{-1} = \sum_{k=0}^{\infty} \frac{(b - a)^k}{(w - a)^{k+1}}, \quad (2.3)$$

and thereupon integrating term by term along Γ_1 .

Show by applying (2.3) to the right-hand side of (2.1) that

$$f(b) = \sum_{k=0}^{\infty} A_k (b - a)^k, \quad |b - a| < r,$$

where

$$A_k = \frac{1}{2\pi i} \int_{\Gamma_1} \frac{f(w)}{(w - a)^{k+1}} dw, \quad k \in \mathbb{N}.$$

2.7. Conclude from Ex. 2.6 that if $\Delta(a; \rho) \subset O$, then

$$f(z) = \sum_0^{\infty} A_k(z-a)^k, \quad z \in \Delta(a; \rho).$$

We see that f is of power series type.

2.8. Show using Theorem 2.2 and Ex. 2.7 that if f is analytic on O , then f is of power series type. Conclude that the C'' hypothesis of Theorem 8.1, Ch. IV, is fulfilled by f that are analytic. Note that the equivalence of *analytic* and *power series type* is now established.

2.9. Given $f: O \rightarrow \mathbb{C}$ where O is an open subset of $\mathbb{C}$ and f has the local primitive property of Ex. 2.4. The continuity of f is not assumed. Show that f is analytic on O .

3. THE CAUCHY–GOURSAT THEOREM FOR A SINGULAR CELL AND SOME CONSEQUENCES

By the rubric “the Cauchy–Goursat theorem for a singular cell” we understand the version of the Cauchy–Goursat theorem that we are about to formulate, the terminology being drawn from algebraic topology. The proof given below is due to F. M. Stewart (cf. [53]).

Theorem 3.1: *Let O be an open subset of $\mathbb{C}$. Let f be analytic on O . Let $S = [0, 1] \times [0, 1]$. Let $\varphi: S \rightarrow O$ be continuous on S . We introduce paths $\gamma_1, \gamma_2, \gamma_3, \gamma_4$ where*

$$\begin{aligned} \gamma_1(t) &= \varphi(t, 0), & \gamma_2(t) &= \varphi(1, t), \\ \gamma_3(t) &= \varphi(1-t, 1), & \gamma_4(t) &= \varphi(0, 1-t), \end{aligned}$$

the domain of each γ_k being $[0, 1]$. Each γ_k is assumed to be rectifiable. Then

$$\sum_1^4 \int_{\gamma_k} f dz = 0.$$

PROOF: We dismiss at once the case where $O = \mathbb{C}$ since f has a primitive in this case. Suppose then that $O \neq \mathbb{C}$. We introduce

$$m = \text{dist}(\mathbb{C} - O, \varphi(S)).^\dagger$$

By Ex. 5.8, Ch. III, $m > 0$. Let n be a positive integer so chosen that $\omega_\varphi(\sqrt{2}/n) < m$. We introduce

$$\begin{aligned} z_{jk} &= \varphi(j/n, k/n), & j, k &= 0, \dots, n; \\ S_{jk} &= [j/n, (j+1)/n] \times [k/n, (k+1)/n], & j, k &= 0, \dots, n-1. \end{aligned}$$

[†] Given $A, B \subset X$ a metric space with metric ρ , by the distance between A and B , denoted $\text{dist}(A, B)$, is meant $\inf \{\rho(a, b): (a, b) \in A \times B\}$.

We have

$$\varphi(S_{jk}) \subset \Delta(z_{jk}; m) \subset O.$$

Now let g_{jk} denote a primitive of $f|_{\Delta(z_{jk}; m)}$. It is trivial that the sum of the quantities A_{jk} , where $j, k = 0, \dots, n-1$, is 0, each being itself 0:

$$\begin{aligned} A_{jk} &= [g_{jk}(z_{j+1,k}) - g_{jk}(z_{jk})] + [g_{jk}(z_{j+1,k+1}) - g_{jk}(z_{j+1,k})] \\ &\quad + [g_{jk}(z_{j,k+1}) - g_{jk}(z_{j+1,k+1})] + [g_{jk}(z_{jk}) - g_{jk}(z_{j,k+1})]. \end{aligned}$$

We now observe the important fact that on $\Delta(z_{jk}; m) \cap \Delta(z_{j+1,k}; m)$, which is a *region*, $g_{jk}(z) - g_{j+1,k}(z)$ is independent of z , $j+1 < n$. A corresponding observation holds for g_{jk} and $g_{j,k+1}$, $k+1 < n$. Hence

$$\begin{aligned} [g_{jk}(z_{j+1,k+1}) - g_{jk}(z_{j+1,k})] + [g_{j+1,k}(z_{j+1,k}) - g_{j+1,k}(z_{j+1,k+1})] \\ = 0, \quad j+1 < n, \end{aligned} \quad (3.1)$$

and

$$\begin{aligned} [g_{jk}(z_{j,k+1}) - g_{jk}(z_{j+1,k+1})] + [g_{j,k+1}(z_{j+1,k+1}) - g_{j,k+1}(z_{j,k+1})] \\ = 0, \quad k+1 < n. \end{aligned} \quad (3.2)$$

Using (3.1) and (3.2) we see that the sum of the A_{jk} is equal to the sum of the following four quantities:

$$\begin{aligned} \sum_0^{n-1} [g_{j0}(z_{j+1,0}) - g_{j0}(z_{j0})] &= \int_{\gamma_1} f dz, \\ \sum_0^{n-1} [g_{n-1,k}(z_{n,k+1}) - g_{n-1,k}(z_{nk})] &= \int_{\gamma_2} f dz, \\ \sum_0^{n-1} [g_{j,n-1}(z_{jn}) - g_{j,n-1}(z_{j+1,n})] &= \int_{\gamma_3} f dz, \\ \sum_0^{n-1} [g_{0k}(z_{0k}) - g_{0k}(z_{0,k+1})] &= \int_{\gamma_4} f dz. \end{aligned}$$

Theorem 3.1 follows.

EXERCISES

3.1. Give a formal proof of the next to the last sentence. To be precise, we note that if $f: E \rightarrow G$ where E is finite and G is an abelian group and φ is a univalent map of $\langle 0, n \rangle$ onto E ($n = -1$ if $E = \emptyset$), then $\sum_0^n f \circ \varphi(k)$ is independent of φ . The common

element in question is the *sum* of f , denoted by $\sum_{x \in E} f(x)$ (or some comparable form of notation). In the present situation we are concerned with $\sum_{0 \leq j, k \leq n-1} A_{jk}$. We have $A_{jk} = \sum_{1 \leq i \leq 4} a_{jk}^i$ where

$$\begin{aligned} a_{jk}^1 &= g_{jk}(z_{j+1,k}) - g_{jk}(z_{jk}), \\ a_{jk}^2 &= g_{jk}(z_{j+1,k+1}) - g_{jk}(z_{j+1,k}), \\ a_{jk}^3 &= g_{jk}(z_{j,k+1}) - g_{jk}(z_{j+1,k+1}), \\ a_{jk}^4 &= g_{jk}(z_{jk}) - g_{jk}(z_{j,k+1}). \end{aligned}$$

Starting with

$$\sum_{0 \leq j, k \leq n-1} A_{jk} = \sum_{1 \leq i \leq 4} \left(\sum_{0 \leq j, k \leq n-1} a_{jk}^i \right),$$

and noting

$$\begin{aligned} \sum_{0 \leq j, k \leq n-1} a_{jk}^1 &= \sum_{0 \leq j \leq n-1} a_{j0}^1 + \sum_{1 \leq k \leq n-1} \left(\sum_{0 \leq j \leq n-1} a_{jk}^1 \right), \\ \sum_{0 \leq j, k \leq n-1} a_{jk}^2 &= \sum_{0 \leq k \leq n-1} a_{n-1,k}^2 + \sum_{0 \leq j \leq n-2} \left(\sum_{0 \leq k \leq n-1} a_{jk}^2 \right), \\ \sum_{0 \leq j, k \leq n-1} a_{jk}^3 &= \sum_{0 \leq j \leq n-1} a_{j,n-1}^3 + \sum_{0 \leq k \leq n-2} \left(\sum_{0 \leq j \leq n-1} a_{jk}^3 \right), \\ \sum_{0 \leq j, k \leq n-1} a_{jk}^4 &= \sum_{0 \leq k \leq n-1} a_{0k}^4 + \sum_{1 \leq j \leq n-1} \left(\sum_{0 \leq k \leq n-1} a_{jk}^4 \right), \end{aligned}$$

draw the desired conclusion by regrouping and using (3.1) and (3.2).

3.2. Prove the following variant of Theorem 3.1. Let O and f be as in Theorem 3.1. Let $\varphi: S \rightarrow O$ be continuous on S and satisfy $\varphi(0, t) = \varphi(1, t)$, $0 \leq t \leq 1$. Let γ_1 and γ_3 be rectifiable. γ_2 and γ_4 need not be. Then

$$\int_{\gamma_1} f dz = \int_{\gamma} f dz,$$

where $\gamma(t) = \gamma_3(1-t)$, $0 \leq t \leq 1$.

We are in a position to draw some important conclusions. Our concern will be with paths. For convenience we make the *normalization* that their domain is $[0, 1]$. Given paths γ_0 and γ_1 in a topological space X which satisfy $\gamma_0(0) = \gamma_1(0)$, $\gamma_0(1) = \gamma_1(1)$, we say that γ_0 is *homotopic to γ_1 relative to X* provided that there exists a continuous map φ of $[0, 1] \times [0, 1]$ into X satisfying:

$$\begin{aligned} \varphi(t, 0) &= \gamma_0(t), & \varphi(t, 1) &= \gamma_1(t), & 0 \leq t \leq 1; \\ \varphi(0, t) &= \gamma_0(0), & \varphi(1, t) &= \gamma_0(1), & 0 \leq t \leq 1. \end{aligned}$$

The following theorem is an immediate consequence of Theorem 3.1.

Theorem 3.2: *Let there be given O an open subset of $\mathbb{C}$, f analytic on O , γ_0 and γ_1 rectifiable paths in O with the same initial points and the same terminal points. If γ_0 is homotopic to γ_1 relative to O , then*

$$\int_{\gamma_0} f dz = \int_{\gamma_1} f dz.$$

Closed curves. Given a topological space X , by a *closed curve* in X is meant a continuous map Γ of $\mathbb{R}$ into X satisfying $\Gamma(t+1) = \Gamma(t)$, $t \in \mathbb{R}$.

Homotopy of closed curves. Given Γ_0 and Γ_1 , closed curves in X a topological space, we say that Γ_0 is (*freely*) *homotopic to Γ_1 relative to X* provided that there exists a continuous map φ of $\mathbb{R} \times [0, 1]$ into X satisfying:

$$\varphi(s+1, t) = \varphi(s, t)$$

for all (s, t) ;

$$\varphi(t, 0) = \Gamma_0(t),$$

$$\varphi(t, 1) = \Gamma_1(t), \quad t \in \mathbb{R}.$$

Suppose that X is a metric space. If $\Gamma|_{[a, a+1]}$ is rectifiable for some a , then it is for all. In this case we term Γ *rectifiable*. The length of $\Gamma|_{[a, a+1]}$ is independent of a . The common value in question is termed the *length of Γ* .

Line integrals. Suppose that Γ is a rectifiable closed curve in $\mathbb{C}$ and that $f: E \rightarrow \mathbb{C}$, $E \subset \mathbb{C}$ and containing the image of Γ . Suppose that $f \circ \Gamma$ is a continuous map of $\mathbb{R}$ into $\mathbb{C}$. Then

$$\int_{\Gamma|_{[a, a+1]}} f dz \tag{3.3}$$

is independent of $a \in \mathbb{R}$. We term the common value of (3.3) *the line integral of f along Γ* and denote it by

$$\int_{\Gamma} f dz. \tag{3.4}$$

In the case of the special closed curve

$$\Gamma(t) = a + re^{2\pi it}, \quad a \in \mathbb{C},$$

r positive, we shall write for (3.4) the conventional

$$\oint_{C(a,r)} f dz,$$

the value of which, it is to be noted, is

$$\int_{\alpha}^{\alpha+2\pi} f(a + re^{i\theta})ire^{i\theta} d\theta, \quad \alpha \in \mathbf{R}.$$

By appeal to Ex. 3.2, this section, we have a Cauchy theorem for freely homotopic closed curves. It states

Theorem 3.3: *Let O and f be as in Theorem 3.2, this section. Let Γ_0 and Γ_1 be rectifiable closed curves in O such that Γ_0 is homotopic to Γ_1 relative to O . Then*

$$\int_{\Gamma_0} f dz = \int_{\Gamma_1} f dz.$$

EXERCISES

3.3. Apply Theorem 3.3 to obtain the results of Exs. 2.5 and 2.6, this chapter, for f analytic *without assuming* the continuity of f' .

3.4. Show that if $|b - a| > r$, then

$$\oint_{C(a;r)} (z - b)^{-1} dz = 0,$$

first with the aid of Theorem 3.3 (deforming $\Gamma_0(t) = a + re^{2\pi it}$ through “concentric circles” to the constant $\Gamma_1(t) = a$), and second using

$$(z - b)^{-1} = -(b - a)^{-1} \sum_0^{\infty} \left(\frac{z - a}{b - a} \right)^k.$$

3.5. Given f analytic at each point of $\overline{\Delta(a; r)}$, $0 < r < +\infty$. Show using the expansion

$$f(z) = \sum_0^{\infty} A_k(z - a)^k, \quad |z - a| < r,$$

where

$$A_k = \frac{1}{2\pi i} \oint_{C(a;r)} \frac{f(w)}{(w - a)^{k+1}} dw, \quad k \in \mathbf{N},$$

that

$$f^{(k)}(a) = \frac{k!}{2\pi i} \oint_{C(a;r)} \frac{f(w)}{(w - a)^{k+1}} dw, \quad k \in \mathbf{N}. \quad (3.5)$$

We shall encounter later more general formulas of this kind.

3.6. Given $a \in \mathbb{C}$, $0 < r_1 < r_2 < +\infty$. Let

$$A = \{z: r_1 \leq |z - a| \leq r_2\}.$$

Let f be analytic at each point of A . Show with the aid of Theorem 3.3 that

$$\oint_{C(a; r_2)} f dz = \oint_{C(a; r_1)} f dz$$

(Cauchy theorem for an annulus) and that

$$f(z) = \frac{1}{2\pi i} \oint_{C(a; r_2)} \frac{f(w)}{w - z} dw - \frac{1}{2\pi i} \oint_{C(a; r_1)} \frac{f(w)}{w - z} dw,$$

$r_1 < |z - a| < r_2$ (Cauchy formula for an annulus). What is the value of the right-hand side when $|z - a| > r_2$ or $|z - a| < r_1$? We shall see in Ch. VI that the Cauchy formula for an annulus has many important applications.

For the Cauchy formula we may apply the Cauchy theorem for an annulus to

$$g(w) = \frac{f(w) - f(z)}{w - z}, \quad w \neq z,$$

$$g(z) = f'(z),$$

and note that we are assured that g is analytic at each point of A .

3.7. Let $\sum_0^\infty a_k z^k$ and $\sum_0^\infty b_k z^k$ converge for $|z| < r$. Let f and g denote their respective sums on $\Delta(0; r)$. Let $h = fg$. Let $h(z) = \sum_0^\infty c_k z^k$, $|z| < r$.† Conclude, interpreting the series as Taylor series of their sums and using the Leibniz formula [cf. p. 202, 32a] for $(fg)^{(k)}$, that $c_k = \sum_0^k a_j b_{k-j}$, $k \in \mathbb{N}$.

3.8. Given f a continuous map of $\overline{\Delta(a; r)}$ into $\mathbb{C}$, analytic in $\Delta(a; r)$, $a \in \mathbb{C}$, $0 < r < +\infty$. Show that

$$\oint_{C(a; r)} f dz = 0,$$

and that

$$\oint_{C(a; r)} f(w) (w - z)^{-1} dw$$

is $2\pi i f(z)$ when $|z - a| < r$ and is 0 when $|z - a| > r$. Hint: Replace r by ρ , $0 < \rho < r$, and consider

$$\lim_r \oint_{C(a; \rho)}$$

with the appropriate integrand.

3.9. Given f analytic on an open subset O of $\mathbb{C}$. Suppose that $|f(z)| \leq M$, $z \in O$,

† Putting aside the developments of Ch. II we are assured that h has such an expansion since it is analytic on $\Delta(0; r)$.

where M is a positive number. Show using the formula for $f'(a)$ of Ex. 3.5 that $|f'(z)| \leq M/d(z)$, where $d(z)$ is the distance between $\{z\}$ and $C - O$. Conclude Liouville's theorem when $O = C$. Show that when

$$z, w \in \overline{\Delta(a; r)} \subset \Delta(a; 2r) \subset O,$$

we have

$$|f(w) - f(z)| \leq \frac{2M|w - z|}{r}.$$

3.10. Let g be a complex-valued continuous function on $C(a; r)$, $a \in C$, $0 < r < +\infty$. Let

$$f(z) = \oint_{C(a; r)} \frac{g(w)}{w - z} dw, \quad z \in C - C(a; r).$$

Show that f is analytic on its domain, that

$$f(z) = \sum_{k=0}^{\infty} A_k(z-a)^k, \quad |z-a| < r,$$

where

$$A_k = \oint_{C(a; r)} \frac{g(w)}{(w-a)^{k+1}} dw, \quad k \in \mathbb{N}, \quad (3.6)$$

and that

$$f(z) = \sum_{k=1}^{\infty} A_{-k}(z-a)^{-k}, \quad |z-a| > r,$$

where $-A_{-k}$ is given by (3.6) with k replaced by $-k$. Hint: For $|z-a| < r$,

$$(w-z)^{-1} = (w-a)^{-1} \sum_{k=0}^{\infty} \left(\frac{z-a}{w-a} \right)^k$$

and for $|z-a| > r$,

$$(w-z)^{-1} = -(z-a)^{-1} \sum_{k=0}^{\infty} \left(\frac{w-a}{z-a} \right)^k.$$

Suppose that h is analytic at each point of $\overline{\Delta(a; r)} - \{a\}$, $0 < r < +\infty$, and is bounded and that a is not in the domain of h . Show that h is the restriction of a function analytic at each point of $\Delta(a; r)$ (Riemann). Hint: Use the Cauchy formula for the annulus, $A = \{s \leq |z-a| \leq r\}$, and consider what happens when s tends to 0. Thereupon apply the first part of this exercise.

3.11. Let O be an open subset of C . Let f_n be analytic on O , $n \in \mathbb{N}$. Suppose that $(f_n)_{n \in \mathbb{N}}$ tends uniformly to f in O . Show with the aid of the Cauchy formula for the

disk and Ex. 3.10 that f is analytic on O . Show with the aid of the formulas (3.5) that $(f_n^{(k)})_{n \in \mathbb{N}}$ tends to $f^{(k)}$ uniformly in O , $k \in \mathbb{N}$.†

The second result of this exercise yields the Weierstrass Double Series Theorem: *Given $r > 0$. Suppose that $\sum_0^\infty a_k^{(n)} z^k$ converges for $|z| < r$ and has sum g_n on $\Delta(0; r)$, $n \in \mathbb{N}$. Let $\sum_0^\infty g_n$ converge uniformly in $\Delta(0; r)$ and let G denote the sum. Let $G(z) = \sum_0^\infty A_k z^k$, $|z| < r$. Then $A_k = \sum_0^\infty a_k^{(n)}$, $k \in \mathbb{N}$, the sum, of course, being with respect to n . Why is the term “double series” applied?*

3.12. The hypotheses of the first two sentences of Ex. 3.11 concerning O and f_n , $n \in \mathbb{N}$, prevail. Suppose that $\sum_0^\infty |f_n|$ converges uniformly in O . Then $\sum_0^\infty |f_n'|$ converges uniformly in O .

3.13. Show that $\sum_0^\infty z^n/(1 - z^n)$ converges uniformly in $\Delta(0; 1)$. Determine the coefficients of the power series representation of its sum and interpret their number-theoretic significance.

3.14. This exercise is intended for experience in computation with power series.

(a) The function given by $\text{Log}(1 + e^z)$, z small, is analytic. Determine the coefficients of its power series expansion as sums of infinite series with the aid of the Double Series Theorem. It will be convenient to determine the coefficients of the power series expansion of $(e^z - 1)^k$, $k \in \mathbb{N}$, using the binomial theorem. (b) Carry out the same program for $\sin[(1 - z)^{-1}]$ with domain $\Delta(0; 1)$.

3.15. We refer to the binomial series and its sum S_z on $\Delta(0; 1)$. Given $a \in \Delta(0; 1)$, let $f(z) = S_z(a)$, $z \in \mathbb{C}$. Show that f is analytic by direct examination of the binomial series. Show that $f' = f'(0)f$ and that

$$f'(0) = \sum_0^\infty (-1)^k (k+1)^{-1} a^{k+1}$$

without appeal to logarithms. Conclude that

$$S_z(a) = \exp \left[z \sum_0^\infty \frac{(-1)^k}{k+1} a^{k+1} \right].$$

3.16. Porter's Example. A power series may have radius of convergence 1 and yet there may exist $\nu: \mathbb{N} \rightarrow \mathbb{N}$ increasing such that the subsequence of partial sums $(s_{\nu(n)})$ converges uniformly in a region containing $\Delta(0; 1)$ properly. Show that the above assertion is established by the following considerations of M. B. Porter: (a) $A = \{|z(1+z)| < 2\}$ is an open set containing $\overline{\Delta(0; 1)} - \{1\}$; (b) $\sum_0^\infty [(z+z^2)/2]^{4n}$ converges uniformly in A ; (c) $\sum_0^\infty [(z+z^2)/2]^{4k}$ is the partial sum with index 2^{2n+1} of a power series having radius of convergence 1. [Actually A is a region.]

The phenomenon of this example, called “overconvergence of power series,” was discovered by Porter. It has been the subject of numerous investigations. Special mention should be made of those of Ostrowski. (For up-to-date references, cf. [11].)

3.17. Let $(\lambda_n)_{n \in \mathbb{N}}$ be an increasing sequence of real numbers, $\lim_{n \rightarrow \infty} \lambda_n = +\infty$. Let

† For this second part of the exercise the uniformity may be established locally with the aid of (3.5) and thereupon for compact subsets of O with the aid of the open covering property. Thus with $\overline{\Delta(b; 2r)} \subset O$, $0 < r < +\infty$, we may apply (3.5) for each $a \in \overline{\Delta(b; r)}$ to $f - f_a$.

$(a_n): \mathbf{N} \rightarrow \mathbf{C}$. (a) Suppose that there exists $w \in \mathbf{C}$ such that $\sum_0^\infty |a_n e^{-\lambda_n w}|$ converges. Show that $\sum_0^\infty a_n e^{-\lambda_n z}$ converges uniformly and absolutely on $\{\operatorname{Re} z \geq \operatorname{Re} w\}$. What can be said about the sum on this set? (b) Suppose that there exists $w \in \mathbf{C}$ such that $\sum_0^\infty a_n e^{-\lambda_n w}$ converges. Let $0 < \alpha < 1$. Show that $\sum_0^\infty a_n e^{-\lambda_n z}$ converges uniformly on $\{\operatorname{Re}(z - w) > \alpha |z - w|\}$ and conclude that $\sum_0^\infty a_n e^{-\lambda_n z}$ converges uniformly in $\{\operatorname{Re} z > \operatorname{Re} w\}$. Hint: Let $b_n = a_n e^{-\lambda_n w}$ and let s satisfy $\operatorname{Re} s > \alpha |s|$. Apply the Abel identity to $\sum_{m+1}^n b_n e^{-\lambda_n s}$ and note that

$$e^{-\lambda_n s} - e^{-\lambda_{n+1} s} = s \int_{\lambda_n}^{\lambda_{n+1}} e^{-st} dt,$$

so that

$$\begin{aligned} |e^{-\lambda_n s} - e^{-\lambda_{n+1} s}| &\leq |s| \int_{\lambda_n}^{\lambda_{n+1}} e^{-(\operatorname{Re} s)t} dt \\ &= \frac{|s|}{\operatorname{Re} s} (e^{-\lambda_n \operatorname{Re} s} - e^{-\lambda_{n+1} \operatorname{Re} s}). \end{aligned}$$

The series considered here are the Dirichlet series (cf. [117] for an introductory account).

3.18. Given O open in $\mathbf{R}^2$ and $(u_n)_{n \in \mathbf{N}}$ where each u_n is harmonic on O . Suppose that (u_n) tends to u uniformly in O . Then u is harmonic in O . Hint: Use the relation between harmonic and analytic functions and formulas (5.6) and (5.7) of Ch. IV.

3.19. Suppose that f is analytic on $\Delta(0; r)$, $0 < r < +\infty$. Suppose further that for each $a \in C(0; r)$ there exists g_a analytic on a disk Δ_a with center a such that

$$f(z) = g_a(z), \quad z \in \Delta(0; r) \cap \Delta_a.$$

Show that

$$f \cup \bigcup_{|a|=r} g_a$$

is an analytic function and hence that the radius of convergence of the power series with sum f on $\Delta(0; r)$ exceeds r . Conclude that if $\sum_0^\infty a_k z^k$ has a finite positive radius of convergence r , there exists $b \in C(0; r)$ such that if $\rho(t)$ denotes the radius of convergence of the power series expansion of $\sum_0^\infty a_k (tb + z)^k$, $\rho(t) = (1 - t)r$, $0 \leq t < 1$.

3.20. Let $f: E \rightarrow \mathbf{C}$ where E is a compact subset of $\mathbf{C}$ containing no isolated points. [$a \in E$ is said to be an *isolated* point of E provided that $a \notin \overline{E - \{a\}}$.] Suppose that for each $a \in E$ there exists g analytic at a such that $g(z) = f(z)$ for $z \in E$ sufficiently near a . Show that there exists h analytic on an open set $O \supset E$ such that $f = h|E$.

3.21. Let O and Ω be open subsets of $\mathbf{C}$. Let $f: O \rightarrow \mathbf{C}$ be continuous on O and let $g: \Omega \rightarrow O$ be analytic on Ω . Let γ be a rectifiable path in Ω . Let $\Gamma = g \circ \gamma$. Show that Γ is rectifiable and that

$$\int_\Gamma f dz = \int_\gamma (f \circ g) g' dz.$$

3.22. Treat Theorem 3.3 of the present section by methods of the calculus under the assumption that φ satisfies appropriate conditions to be formulated and stated;

that is, study the dependence on t of

$$\int_{\Gamma_t} f dz$$

where $\Gamma_t(s) = \varphi(s, t)$, $0 \leq s \leq 1$.

Further theorems of Cauchy type. From the point of view of applications it is desirable to have available certain special versions of the Cauchy theorem and formula. We treat the Cauchy theorem and formula for a *strip*.

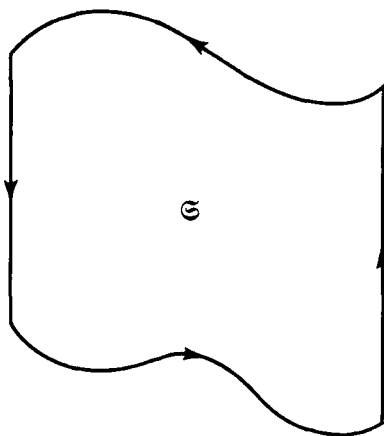


FIG. 5.2

Let $Y_1, Y_2: [a, b] \rightarrow \mathbf{R}$, where $-\infty < a < b < +\infty$, each Y_k is continuous on $[a, b]$ and $V[Y_k] < +\infty$, $k = 1, 2$. We suppose that $Y_1(x) < Y_2(x)$, $a < x < b$. By $s(x)$ we denote the segment

$$[x + i Y_1(x), x + i Y_2(x)].$$

By the strip $\mathfrak{S}$ determined by (Y_1, Y_2) is meant $\bigcup_{a \leq x \leq b} s(x)$. By the *positively sensed boundary path* of $\mathfrak{S}$ is meant the path with domain $[0, 4]$ given by

$$\begin{aligned} B(t) &= a(1-t) + bt + iY_1[a(1-t) + bt], & 0 \leq t \leq 1; \\ &= b + i[(2-t)Y_1(b) + (t-1)Y_2(b)], & 1 < t \leq 2; \\ &= b(3-t) + a(t-2) + iY_2[b(3-t) + a(t-2)], & 2 < t \leq 3; \\ &= a + i[(4-t)Y_2(a) + (t-3)Y_1(a)], & 3 < t \leq 4. \end{aligned}$$

We note that

$$\begin{aligned}\varphi(s, t) &= a(1 - s) + bs + i\{(1 - t)Y_1[a(1 - s) + bs] \\ &\quad + tY_2[a(1 - s) + bs]\}\end{aligned}$$

maps the unit square $[0, 1] \times [0, 1]$ onto $\mathfrak{S}$. By the Cauchy theorem for the singular cell we see that if f is analytic on an open subset O of $\mathbb{C}$ and $\mathfrak{S} \subset O$, then

$$\int_B f dz = 0. \quad (3.7)$$

Suppose that c is an *interior point* of $\mathfrak{S}$ (that is, $\mathfrak{S}$ is a neighborhood of c ; this is so if and only if $a < \operatorname{Re} c < b$ and $Y_1(\operatorname{Re} c) < \operatorname{Im} c < Y_2(\operatorname{Re} c)$). We have, by proceeding as we did in deriving the Cauchy formula for the disk,

$$f(c) \int_B (z - c)^{-1} dz = \int_B f(z) (z - c)^{-1} dz. \quad (3.8)$$

Now the integral on the left of (3.8) may be evaluated as follows. From (3.7) if $\mathfrak{S}_1$ is a strip *lying over* $\mathfrak{S}$ in the sense that its Y_1 is the Y_2 of $\mathfrak{S}_1$ and B_1 is the positively sensed boundary path of $\mathfrak{S}_1$, then

$$\int_{B_1} (z - c)^{-1} dz = 0,$$

$\mathfrak{S}_1$ lying in $\mathbb{C} - \{c\}$, on which $(z - c)^{-1}$ is analytic. A similar remark holds if $\mathfrak{S}_1$ is a strip *lying under* $\mathfrak{S}$, that is, having as its Y_2 the Y_1 of $\mathfrak{S}$. Let us take a strip lying over $\mathfrak{S}$ having constant Y_2 and a strip lying under $\mathfrak{S}$ having constant Y_1 and sum the integrals of $(z - c)^{-1}$ along the positively sensed boundary paths of these strips and of $\mathfrak{S}$. Taking the constant value of Y_2 for the strip lying over $\mathfrak{S}$ as $\operatorname{Im} c + s$ and the constant value of Y_1 for the strip lying under $\mathfrak{S}$ as $\operatorname{Im} c - s$, where s is large and positive, and noting that the integrals of $(z - c)^{-1}$ along

$$\overrightarrow{(b + i(\operatorname{Im} c + s))(a + i(\operatorname{Im} c + s))}$$

and

$$\overrightarrow{(a + i(\operatorname{Im} c - s))(b + i(\operatorname{Im} c - s))}$$

are small for s large, we see that

$$\int_B (z - c)^{-1} dz = \lim_{+\infty} \left[\int_{\gamma(s)} (z - c)^{-1} dz + \int_{\Gamma(s)} (z - c)^{-1} dz \right],$$

where

$$\gamma(s) = \overrightarrow{(b + i(\operatorname{Im} c - s))(b + i(\operatorname{Im} c + s))}$$

and

$$\Gamma(s) = \overrightarrow{(a + i(\operatorname{Im} c + s))(a + i(\operatorname{Im} c - s))}.$$

Parametrizing $\{\operatorname{Re} z = b\}$ by

$$c + (b - \operatorname{Re} c)(1 + i \tan \theta), \quad -\pi/2 < \theta < \pi/2,$$

we find

$$\lim_{+\infty} \int_{\gamma(s)} (z - c)^{-1} dz = \lim_{\pi/2} \int_{-\varphi}^{\varphi} i(1 - i \tan \theta) d\theta = \pi i,$$

and by an obvious paraphrase that

$$\lim_{+\infty} \int_{\Gamma(s)} (z - c)^{-1} dz = \pi i.$$

Hence

$$\int_B (z - c)^{-1} dz = 2\pi i,$$

and we conclude from (3.8) the *Cauchy formula for a strip*:

$$f(c) = \frac{1}{2\pi i} \int_B f(z) (z - c)^{-1} dz,$$

c an interior point of $\mathfrak{S}$.

It is obvious that we may treat in the same way strips whose generating segments have any given direction and are not necessarily vertical. The following two examples yield useful strips.

(a) Given $0 < r < +\infty$, we take $a = -r$, $b = r$, $Y_1(x) = 0$, $Y_2(x) = \sqrt{(r^2 - x^2)}$. We obtain a (closed) *semicircular disk*.

(b) Let $0 < r_1 < r_2 < +\infty$. Taking $a = -r_2$, $b = r_2$, $Y_1(x) = 0$ when $r_1 \leq |x| \leq r_2$ and $Y_1(x) = \sqrt{(r_1^2 - x^2)}$ when $|x| < r_1$, and $Y_2(x) = \sqrt{(r_2^2 - x^2)}$ we obtain a (closed) *indented semicircular disk*.

EXERCISES

3.23. Starting from the Cauchy formula for a semicircular disk show that if f is analytic at each point of $\{\operatorname{Im} z \geq 0\}$ and $\lim_{\infty} f = 0$, then

$$f(z) = \lim_{+\infty} \frac{1}{2\pi i} \int_{-r}^r f(t) (t-z)^{-1} dt, \quad \text{Im } z > 0.$$

3.24. Demonstrate the following lemma:

$$\lim_{+\infty} \int_0^{\pi/2} e^{-r \sin \theta} d\theta = 0.$$

It is convenient to show first that $\sin \theta \geq 2\theta/\pi$, $0 \leq \theta \leq \pi/2$. This inequality may be treated by showing that with $g(\theta) = \sin \theta - 2\theta/\pi$, we have $g''(\theta) < 0$, $0 < \theta < \pi/2$, and thereupon that the minimum of g on $[0, \pi/2]$ is 0.

This exercise is meant to be ancillary to the following one.

3.25. Calculate

$$\int_0^{+\infty} x^{-1} \sin x \, dx$$

by applying the Cauchy theorem for the *indented semicircular disk* to e^{iz}/z . Let the outer radius tend to $+\infty$ and the inner to 0. The lemma of Ex. 3.24 controls the limiting behavior of the contribution associated with the semicircular arc having the larger radius. The contribution associated with the other semicircular arc is controlled by the fact that $e^{iz}/z = 1/z + f(z)$ where f is analytic on $\mathbb{C}$. Consider also

$$\int_0^{+\infty} x^{-1} (\sin x)^{2n+1} \, dx, \quad n \in \mathbb{N}.$$

3.26. Let $\gamma(\theta) = re^{i\theta}$, $0 \leq \theta \leq \pi/4$, where r is positive. Show that

$$\int_{(0)(re^{i\pi/4})}^{\rightarrow} e^{iz^2} \, dz = \int_0^r e^{ix^2} \, dx + \int_{\gamma} e^{iz^2} \, dz,$$

and conclude that

$$\int_0^{+\infty} \cos(x^2) \, dx = \int_0^{+\infty} \sin(x^2) \, dx = 2^{-1/2} \int_0^{+\infty} e^{-x^2} \, dx.$$

The first two integrals on the preceding line are the so-called Fresnel integrals, the last is the Gauss integral. For a complex-function-theoretic evaluation of the Gauss integral cf. §1, Ch. XIV of this book where the evaluation due to Mordell is given.

3.27. Let f be analytic at each point of $\{\text{Im } z \geq 0\}$. Suppose that

$$\lim_{+\infty} r \int_0^{\pi} |f(re^{i\theta})| \, d\theta = 0.$$

Show that if $0 < \alpha < +\infty$, then

$$\lim_{+\infty} \int_{-r}^r f(x) e^{i\alpha x} \, dx = 0.$$

3.28. Let $0 < r < +\infty$. Consider the sum of the integrals of e^{-z^2} along $\overrightarrow{(-r)(r)}$,

$\overrightarrow{(r)(r+iy)}, \overrightarrow{(r+iy)(-r+iy)}, \overrightarrow{(-r+iy)(-r)}$, and determining the limit of this sum as r tends to $+\infty$ calculate

$$\int_{-\infty}^{+\infty} e^{-x^2} \cos(2xy) dx.$$

3.29. Ankeny's proof of the fundamental theorem of algebra. Let

$$P(z) = z^{n+1} + \sum_0^n a_k z^k$$

and suppose that $P(z)$ is never zero. Introduce

$$Q(z) = z^{n+1} + \sum_0^n \bar{a}_k z^k.$$

$Q(z)$ is never zero. Apply the Cauchy theorem for a semicircular disk to $1/PQ$ and let the radius tend to $+\infty$ obtaining

$$\int_{-\infty}^{+\infty} |P(x)|^{-2} dx = 0.$$

What contradiction do we have?

3.30. A theorem of Fejér and F. Riesz. Let $f: \overline{\Delta(0; 1)} \rightarrow \mathbb{C}$ be continuous on $\overline{\Delta(0; 1)}$ and analytic in $\Delta(0; 1)$. Then

$$2 \int_{-1}^1 |f(x)|^2 dx \leq \int_0^{2\pi} |f(e^{i\theta})|^2 d\theta.$$

Hint: Let $g(z) = \overline{f(\bar{z})}$ and apply the Cauchy theorem for the semicircular disk to fg .

Some remarks on Cauchy theorems and formulas. Let $n \in \mathbb{N}$. In case $n \geq 1$ let $a_1, \dots, a_n$ and h satisfy $0 < h < 1$, $-1 < a_1 - h < a_n + h < 1$, and $a_j + h < a_{j+1} - h$, $j = 1, \dots, n-1$. Let E be the set

$$[-1, 1] \times [-1, 1] - \bigcup_{1 \leq j \leq n}]a_j - h, a_j + h[\times]-h, h[.$$

Let $\varphi: E \rightarrow \mathbb{C}$, O being an open subset of $\mathbb{C}$ and φ being continuous on E . Let f be analytic on O . Further let $\beta = \varphi \circ B$ where B is the positively sensed boundary path of $[-1, 1] \times [-1, 1]$, and let $\beta_j = \varphi \circ B_j$, where B_j is the positively sensed boundary path of $[a_j - h, a_j + h] \times [-h, h]$, $j = 1, \dots, n$. Here we borrow the terminology introduced for strips in C.

The following version of the Cauchy theorem holds:

If β and β_j , $j = 1, \dots, n$, are rectifiable, then

$$\int_{\beta} f dz - \sum_1^n \int_{\beta_j} f dz = 0.$$

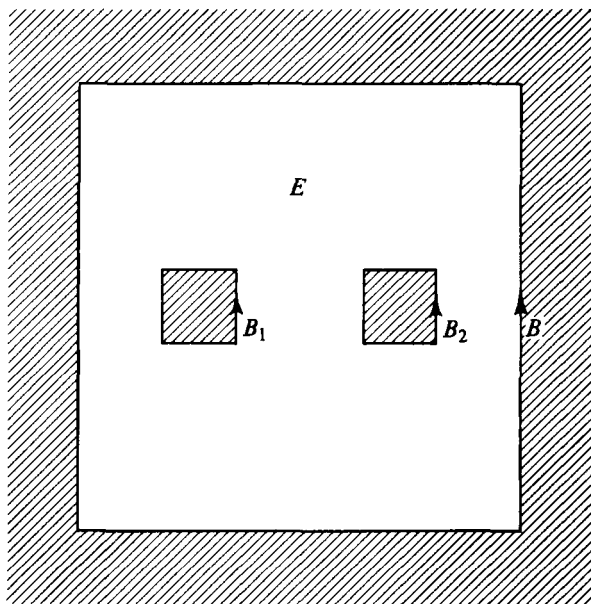


FIG. 5.3

The proof is a direct paraphrase of Theorem 3.1, this section. It suffices to subdivide E into rectangles by the horizontal lines $\{(x, y): y = h\}$, $\{(x, y): y = -h\}$, and the verticals $\{(x, y): x = a_j - h\}$, $\{(x, y): x = a_j + h\}$, $j = 1, \dots, n$, and to subdivide the rectangles obtained into subrectangles following the argument of Theorem 3.1.

A Cauchy formula holds when φ is univalent:

$$f(z) = \frac{\varepsilon}{2\pi i} \left[\int_{\beta} \frac{f(w)}{w - z} dw - \sum_1^n \int_{\beta_j} \frac{f(w)}{w - z} dw \right],$$

z an interior point of $\varphi(E)$. Here ε is independent of z and takes the value 1 or the value -1 . We are stating this result without proof. As may readily be seen, the proof hinges on the calculation of

$$\int_{\beta} (w - z)^{-1} dw - \sum_1^n \int_{\beta_j} (w - z)^{-1} dw.$$

This question is accessible after the proof of the Jordan curve theorem based on the order of a point with respect to a closed curve. The order of a point with respect to a curve admits evaluation in the rectifiable case as a line integral with an integrand of the indicated form (cf. §6, this chapter). We note also without proof that the $\varphi(E)$ consist of the closures of the bounded

regions in $\mathbb{C}$ whose frontiers consist of $n + 1$ disjoint homeomorphs of $C(0; 1)$.†

A large part of the treatment of these questions belongs to the realm of plane topology. The proofs are not trivial. The degree of generality achieved goes far beyond the needs of an introductory course. The interested reader will find pp. 146–149 and 154–155 of [53] pertinent and also the developments of [92].

4. SIMPLE CONNECTIVITY, PRIMITIVES

Among arcwise connected topological spaces the homotopically simply-connected play a distinguished role. The notion admits a number of equivalent formulations. We shall take one that is simple to use and not hard to test in elementary standard situations. We say that an arcwise connected topological space X is *homotopically simply connected* provided that, given paths γ_0, γ_1 in X with domain $[0, 1]$ having a common initial point and a common terminal point, γ_0 is homotopic to γ_1 relative to X .‡ For the case, which interests us especially, of a region in $\mathbb{C}$, we see that *convexity* is a sufficient condition for homotopic simple connectivity. Indeed, if γ_0 and γ_1 are paths with domain $[0, 1]$ in a convex region $A \subset \mathbb{C}$ which have common initial points and common terminal points, the map

$$\varphi(s, t) = (1 - t) \gamma_0(s) + t \gamma_1(s), \quad 0 \leq s, t \leq 1,$$

shows that γ_0 is homotopic to γ_1 relative to A . Hence A is homotopically simply connected. In particular, since a disk $\Delta(a; r)$ and a half-plane $\{\operatorname{Re}(uz) > b\}$, $u \in C(0; 1)$, $b \in \mathbb{R}$, are convex, they are homotopically simply connected.

From the point of view of analytic functions on homotopically simply connected regions the following theorem is fundamental.

Theorem 4.1: *Given f analytic on a homotopically simply connected region $A \subset \mathbb{C}$, there exists g analytic on A such that $g' = f$. That is, f possesses a primitive.*

PROOF: We fix $a \in A$ and let Γ denote the class of rectifiable paths γ in A with domain $[0, 1]$ and initial point a . It is easy to see that

$$A = \{\gamma(1) : \gamma \in \Gamma\}. \quad (4.1)$$

† A homeomorph of a topological space is its image with respect to a homeomorphism.

‡ Other equivalent formulations may be given in terms of the fundamental group of X (cf. [2]).

Indeed, the argument that was used to show that a region in $\mathbf{C}$ is arcwise connected shows that, given $b \in A$, there exists a rectifiable path γ in A with initial point a and terminal point b . We propose

$$g = \left\{ \left(\gamma(1), \int_{\gamma} f dz \right) : \gamma \in \Gamma \right\}.$$

By the homotopic version of the Cauchy integral theorem and (4.1) we see that g is a function with domain A . Further, given $z \in \Delta(b; r) \subset A$, we see that

$$g(z) - g(b) = \int_{\vec{bz}} f(w) dw$$

and we conclude that $g'(b) = f(b)$ (cf. the proof of Theorem 2.2, this chapter). Theorem 4.1 follows.

EXERCISES

4.1. Show that the conclusion of Theorem 4.1 persists when the following condition replaces homotopic simple connectivity: there exists $a \in A$ such that all paths γ in A with domain $[0, 1]$ satisfying $\gamma(0) = \gamma(1) = a$ are homotopic to the constant path $\gamma_1(t) = a$, $0 \leq t \leq 1$, relative to A . Actually this condition is equivalent to homotopic simple connectivity.

4.2. Show that $f: O \rightarrow \mathbf{C}$, where O is an open subset of $\mathbf{C}$, possesses a primitive if and only if f is analytic on O and

$$\int_{\gamma} f dz = 0$$

for every closed rectifiable curve γ in O .

***4.3.** Let Ω be a homotopically simply connected region in $\mathbf{R}^2$. Let $g, h \in C'(\Omega)$ and satisfy $g_y = h_x$. Show that there exists $f \in C'(\Omega)$ such that $f_x = g$, $f_y = h$. It is suggested that Theorem 4.1, this section, serve as a model, the line integrals being of the form

$$\int_{\gamma} g dx + h dy,$$

where the γ are rectifiable paths in Ω issuing from a given point of Ω . Of course, the essential definitions concerning line integrals of the indicated form are wanted. This exercise is intended for readers with the necessary background.

Holomorphic simple connectivity. A region in $\mathbf{C}$ is said to be *holomorphically simply connected* provided that every function analytic on it possesses a primitive. Theorem 4.1 says that a region in $\mathbf{C}$ that is homotopically simply connected is holomorphically simply connected. The converse is true, but it is

not so immediate. A proof of the converse may be based on the Riemann mapping theorem (cf. §1, Ch. XVII of this book), which may be stated: if A is a holomorphically simply connected region which is a proper part of $\mathbb{C}$, there exists a univalent analytic function with domain A and image $\Delta(0; 1)$; and on the fact that the inverse of a univalent analytic function is analytic (for the latter theorem, cf. Theorem 9.1, this chapter).

The following theorem is useful for establishing holomorphic simple connectivity.

Theorem 4.2: *Let A and B be holomorphically simply connected regions in $\mathbb{C}$ with $A \cap B \neq \emptyset$, connected. Then $A \cup B$ is holomorphically simply connected.*

PROOF: $A \cup B$ is a region. Suppose that f is analytic on $A \cup B$. Let $a \in A \cap B$; let g denote the primitive of $f|_A$ satisfying $g(a) = 0$; let h denote the primitive of $f|_B$ satisfying $h(a) = 0$. Then $g(z) = h(z)$, $z \in A \cap B$, since $g(a) = h(a)$ and $g'(z) = f(z) = h'(z)$. Hence $g \cup h$ is a primitive of f . The assertion follows.

Examples. $\mathbb{C} - \{a + t: t \in \mathbb{R}^+\}$, $a \in \mathbb{R}$, is holomorphically simply connected. It admits the representation

$$[\{\operatorname{Im} z < 0\} \cup \{\operatorname{Re} z < a\}] \cup \{\operatorname{Im} z > 0\}. \quad (4.2)$$

Similarly, $\mathbb{C}$ less two slits, one $\{a + t: t \in \mathbb{R}^+\}$, the other $\{b - t: t \in \mathbb{R}^+\}$, $b < a$, is holomorphically simply connected. To see this it suffices to replace $\{\operatorname{Re} z < a\}$ by $\{b < \operatorname{Re} z < a\}$ in (4.2).

EXERCISES

4.4. Show that $\mathbb{C} - \{0\}$ is not holomorphically simply connected by calculating

$$\oint_{C(0;1)} z^{-1} dz.$$

4.5. Let α denote the primitive of $(1 + z^2)^{-1}$ on $A = \mathbb{C}$ less the slits

$$\{i(t + 1): t \in \mathbb{R}^+\} \quad \text{and} \quad \{-i(t + 1): t \in \mathbb{R}^+\}$$

satisfying $\alpha(0) = 0$. Let $f(z) = \tan[\alpha(z)]$ at the points $z \in A$ for which $\alpha(z)$ is not of the form $(n + \frac{1}{2})\pi$, $n \in \mathbb{Z}$. Show that

$$\frac{f'(z)}{1 + [f(z)]^2} = \frac{1}{1 + z^2}$$

for z in the domain of f and conclude that $\alpha[f(z)] = \alpha(z)$ for z small and thereupon that f is the identity map of A onto itself.

5. ANALYTIC AND CONTINUOUS LOGARITHMS AND POWERS

Let f be analytic on an open set $O \subset \mathbb{C}$ and such that $f(z) \neq 0$ for all $z \in O$. We seek a necessary and sufficient condition for f to possess an analytic logarithm. Suppose that g is an analytic logarithm of f . From $f = \exp \circ g$ we obtain, using the chain rule, $f' = fg'$. Hence a *necessary* condition for f to possess an analytic logarithm is that f'/f possess a primitive. This condition is *sufficient*. For suppose now that g is a primitive of f'/f . We see that $[f \exp \circ (-g)]'$ is the constant zero and hence $f \exp \circ (-g) = c$, where c is constant on each component of O , that is, the restriction of c to a component is constant. Let k be constant on each component of O and satisfy $c = \exp \circ k$. Then $g + k$ is an analytic logarithm of f . To sum up we have

Theorem 5.1: *Given f analytic on an open set $O \subset \mathbb{C}$ such that $f(z) \neq 0$, $z \in O$. Then a necessary and sufficient condition that f possess an analytic logarithm is that f'/f possess a primitive.*

From our earlier study of logarithms and powers we see now that if f as above possesses an analytic logarithm and h is analytic on the domain of f , then there exists an analytic h th power of f .

In case O is a holomorphically simply connected region we are guaranteed that, for every f analytic on O not taking the value 0, the logarithmic derivative f'/f admits a primitive, and hence f has an analytic logarithm. It is to be noted that this assertion appeals to the analyticity of f' .

EXERCISES

5.1. Let f be a continuous map of a topological space X into $\mathbb{C} - \{0\}$. Suppose that g is a continuous logarithm of f . Suppose that $a \in X$ has the property that for each neighborhood U of a there exists a connected neighborhood of a contained in U . Then there exists a neighborhood V of a and an analytic logarithm L of the identity map on $\Delta[f(a); |f(a)|]$ such that $g(x) = L[f(x)]$, $x \in V$.

5.2. Let $\sum_{k=0}^{\infty} a_k z^k = \text{Log}(1 + e^z)$, z small. Let ρ denote the radius of convergence of $\sum_{k=0}^{\infty} a_k z^k$. Show that the sum of the series on $\Delta(0; \rho)$ is an analytic logarithm of $1 + e^z$ restricted to $\Delta(0; \rho)$ and determine ρ explicitly from properties of the exponential function.

5.3. Show that $1 - z^2$, $z \in \mathbb{C} - [-1, 1]$, possesses a unique analytic square root S satisfying $\lim_{\infty} iz^{-1} S(z) = 1$ and show by the Cauchy integral theorem and formula for a semicircle with two indentations that

$$\int_{-1}^1 \frac{dx}{(x-a)\sqrt{(1-x^2)}} = \pi i [S(a)]^{-1}, \quad a \in \mathbb{C} - [-1, 1].$$

The usual convention of real analysis applies to $\sqrt{(1-x^2)}$.

5.4. Let X be a connected topological space and let $\alpha \in \mathbb{C}$. Show that the continuous α th powers of the constant 1 on X are the constant functions taking the values $e^{2\pi i \alpha k}$, $k \in \mathbb{C}$.

We now consider continuous logarithms in situations where the domain has a special simple character and we shall obtain the essential apparatus to treat the notion of the *topological index* (= *order of a point with respect to a curve*) which enters very naturally into the Cauchy theory as we shall soon see. Suppose then that f is a continuous map of $[a, b]$ into $\mathbb{C} - \{0\}$, $-\infty < a < b < +\infty$. We wish to show that f possesses a continuous logarithm. We proceed as follows. Let $m = \min_{[a, b]} |f|$. Clearly $m > 0$. Now let n denote a positive integer satisfying $\omega_f[n^{-1}(b-a)] < m$. Let

$$x_k = a + n^{-1}k(b-a), \quad k = 0, \dots, n.$$

Since

$$|f(x) - f(x_k)| < m \leq |f(x_k)|, \quad x_k \leq x \leq x_{k+1}, \quad k = 0, \dots, n-1,$$

we see that

$$f(x) \in \Delta[f(x_k); |f(x_k)|], \quad x_k \leq x \leq x_{k+1}.$$

The composition of a continuous logarithm of the identity map on $\Delta[f(x_k); |f(x_k)|]$ with $f|_{[x_k, x_{k+1}]}$ yields a continuous logarithm l_k of $f|_{[x_k, x_{k+1}]}$. We notice that $l_{k+1}(x_{k+1}) - l_k(x_{k+1})$ is an integer multiple of $2\pi i$, $k = 0, \dots, n-2$. Starting with l_0 fixed we adjust the successive l_k to be concordant by adding appropriate constant integer multiples of $2\pi i$ so that the resulting l_k satisfy $l_{k+1}(x_{k+1}) = l_k(x_{k+1})$, $k = 0, \dots, n-2$. Thus l_1 is replaced by $l_1 + [l_0(x_1) - l_1(x_1)]$, etc. We see that with the l_k so modified $\bigcup_{0 \leq k \leq n-1} l_k$ is a continuous logarithm of f .

Now that the case of the segment has been treated, it is easy to treat the rectangle and certain important unbounded domains. Let f now be a continuous map of $[a, b] \times [c, d]$ into $\mathbb{C} - \{0\}$, where a, b are as above and $-\infty < c < d < +\infty$. The procedure is the same as that for the case of a segment up to the point of adjusting the individual continuous logarithms associated with the subrectangles. Thus we suppose that a positive integer n has been found such that

$$f|_{[x_j, x_{j+1}] \times [y_k, y_{k+1}]}$$

has a continuous logarithm l_{jk} , $0 \leq j, k \leq n-1$. Here $x_j = a + jn^{-1}(b-a)$, $y_k = c + kn^{-1}(d-c)$, $j, k = 0, \dots, n$. We note that

$$l_{jk}(x_{j+1}, y) = l_{j+1,k}(x_{j+1}, y) + 2\pi i m(j, k),$$

where $m(j, k)$ is an integer, $y_k \leq y \leq y_{k+1}$. Hence $l_{0k}, \dots, l_{n-1,k}$ may be

rendered concordant so that

$$\bigcup_{0 \leq j \leq n-1} l_{jk}$$

is a continuous logarithm L_k of $f| [a, b] \times [y_k, y_{k+1}]$. This time

$$L_k(x, y_{k+1}) = L_{k+1}(x, y_{k+1}) + 2\pi i \mu(k), \quad a \leq x \leq b,$$

where $\mu(k)$ is an integer, $k = 0, \dots, n-2$. Hence $L_0, \dots, L_{n-1}$ may be rendered concordant so that

$$\bigcup_{0 \leq k \leq n-1} L_k$$

is a continuous logarithm of f .

Two other important cases are those where the domain of the function in question is either $\mathbf{R}$ or $\mathbf{R} \times [0, 1]$. Suppose that f is a continuous map of $\mathbf{R}$ into $\mathbf{C} - \{0\}$. Let α denote a given logarithm of $f(0)$. For each positive integer n let l_n denote the continuous logarithm of $f| [-n, n]$ satisfying $l_n(0) = \alpha$. Then, as is easily seen,

$$\bigcup_{n \in \mathbf{N}} l_{n+1}$$

is a continuous logarithm of f . The same argument holds when the domain is $\mathbf{R} \times [0, 1]$, the normalization now being made at $(0, 0)$ and $[-n, n] \times [0, 1]$ replacing $[-n, n]$.

Application to closed curves. We recall that the existence of continuous logarithms implies the existence of continuous arguments. Let γ be a closed curve and let a be a point of $\mathbf{C}$ not on γ . Let θ be a continuous argument of $\gamma - a$. Because of the fact that $\gamma(t+1) = \gamma(t)$, $t \in \mathbf{R}$, we see that

$$\frac{\theta(t+1) - \theta(t)}{2\pi} \quad (5.1)$$

is an integer independent of t . It is also independent of admitted θ . We term the common value of (5.1) *the order of a with respect to γ or the topological index of γ at a* and denote it by $O(a; \gamma)$.

Let us now see that $O(a; \gamma)$ is a *homotopy invariant*; that is, when γ_0 is homotopic to γ_1 relative to $\mathbf{C} - \{a\}$, then $O(a; \gamma_0) = O(a; \gamma_1)$. For this purpose suppose that f is a continuous map of $\mathbf{R} \times [0, 1]$ into $\mathbf{C} - \{0\}$ satisfying $f(x+1, y) = f(x, y)$, $(x, y) \in \mathbf{R} \times [0, 1]$. Let φ denote a continuous argument of f . Then

$$\frac{\varphi(x+1, y) - \varphi(x, y)}{2\pi} = n(f)$$

is an integer independent of (x, y) . If we now consider closed curves γ_0 and γ_1 where γ_0 is homotopic to γ_1 relative to $\mathbf{C} - \{a\}$, we see that there exists f of the stated type such that $f(t, 0) + a = \gamma_0(t)$, $f(t, 1) + a = \gamma_1(t)$, $t \in \mathbf{R}$. We have

$$O(a; \gamma_0) = n(f) = O(a; \gamma_1)$$

and the asserted invariance follows.

Suppose now that $b \in \Delta(a; r) \subset \mathbf{C} - \gamma$. We define f on $\mathbf{R} \times [0, 1]$ by

$$f(s, t) = \gamma(s) - [(1 - t)a + tb],$$

and considering a continuous argument of f we conclude that $O(a; \gamma) = O(b; \gamma)$. By the standard connectedness argument we see that $O(z; \gamma)$ is independent of z on a component of $\mathbf{C} - \gamma$. We shall be more precise about the unbounded component. For if $|a| > \max_{\mathbf{R}} |\gamma|$, noting that

$$\gamma(t) - a = (-a) \left[1 - \frac{\gamma(t)}{a} \right], \quad t \in \mathbf{R},$$

we see that $O(a; \gamma) = 0$ since $\text{Log}[1 - \gamma(t) a^{-1}]$ is a continuous logarithm of $1 - \gamma(t) a^{-1}$ left unaltered by the translation taking t into $t + 1$.

In §6, this chapter, we shall see the importance of the notion of the order of a point for the Cauchy theory.

EXERCISES

5.5. Let f be a continuous map of $[a, b]$ into $\mathbf{C} - \{0\}$, $-\infty < a < b < +\infty$. Let $(t_k)_n$ be the increasing sequence whose image is a given partitioning of $[a, b]$. Let α denote a continuous argument of f and let α_k denote a continuous argument of $f|_{[t_k, t_{k+1}]}$, $k = 0, \dots, n-1$. Show that

$$\alpha(b) - \alpha(a) = \sum_{k=0}^{n-1} [\alpha_k(t_{k+1}) - \alpha_k(t_k)].$$

This simple rule is very convenient for applications.

5.6. Let $\gamma(t) = \exp(2\pi it)$, $t \in \mathbf{R}$. Show that $O(0; \gamma) = 1$ and conclude that $\mathbf{C} - \{0\}$ is not homotopically simply connected.

5.7. Let $A = \{r_1 < |z| < r_2\}$, $0 \leq r_1 < r_2 \leq +\infty$. Show that if γ_0 and γ_1 are closed curves in A , then γ_0 is homotopic to γ_1 relative to A if and only if $O(0; \gamma_0) = O(0; \gamma_1)$. Consider the corresponding question for paths in A which have a given point $a \in A$ as both initial point and terminal point.

5.8. A lemma on order. Let γ be a closed curve in $\mathbf{C} - \{0\}$. Suppose that $\text{Re } \gamma$ takes the value 0 only a finite number of times on $[0, 1]$. $\text{Re } \gamma$ is said to *change sign at* t provided that

$$\text{sg}[\text{Re } \gamma(t + \varepsilon) \text{Re } \gamma(t - \varepsilon)] = -1$$

for ε small and positive. Show that $2|O(0; \gamma)|$ does not exceed the number of points on $[0, 1[$ at which $\operatorname{Re} \gamma$ changes sign (R. J. Backlund).

5.9. Determine with the aid of the results of this section the continuous (respectively, analytic) maps f of $\mathbb{C}$ into itself satisfying $f(z + w) = f(z)f(w)$, $z, w \in \mathbb{C}$; of $\mathbb{C} - \{0\}$ into itself satisfying $f(zw) = f(z)f(w)$, $z, w \in \mathbb{C} - \{0\}$. Hint: For the first part consider a continuous logarithm of f in the nontrivial case, for the second part consider $f \circ \exp$.

5.10. Let f be a continuous map of $\overline{\Delta(0; 1)}$ into itself. Show that there exists $a \in \Delta(0; 1)$ satisfying $f(a) = a$. [This is a special case of the Brouwer fixed point theorem.] Hint: Suppose the assertion false and consider $O(0; \gamma_t)$ and $O(0; \Gamma_t)$, $0 \leq t \leq 1$, where

$$\begin{aligned}\gamma_t(s) &= f(te^{2\pi is}) - te^{2\pi is}, \\ \Gamma_t(s) &= tf(e^{2\pi is}) - e^{2\pi is}, \quad s \in \mathbb{R}.\end{aligned}$$

This ingenious argument is due to A. P. Morse.

5.11. Let $-\infty < a < b < +\infty$. Let A and B be continuous real-valued functions with domain $[a, b]$. Let $w: [a, b] \rightarrow \mathbb{C} - \{0\}$ be such that w'' exists and w satisfies $w'' + Aw' + Bw = 0$. Let θ denote a continuous argument of w . Show that if θ is not constant, then either $\theta(s) < \theta(t)$ for all s, t satisfying $a \leq s < t \leq b$, or $\theta(s) > \theta(t)$ for all s, t satisfying $a \leq s < t \leq b$.

6. CERTAIN INTEGRALS REPRESENTING ANALYTIC FUNCTIONS

The Cauchy integral formulas are special instances of integrals representing analytic functions. As a preliminary to the study of such integrals we first consider functions defined on a product set which are “analytic in the first variable and are continuous in both.” Such functions were examined systematically in a euclidean setting by Bôcher [136] who termed them *semianalytic*. We shall see that they have interesting properties.

To be specific, let O be an open subset of $\mathbb{C}$ and let X be a compact metric space. Let F be a continuous map of $O \times X$ into $\mathbb{C}$ and for each $p \in X$ let the function f_p assigning the value $F(z, p)$ to $z \in O$ be analytic on O . We define G on $O \times O \times X$ by

$$\begin{aligned}G(z, w, p) &= \frac{F(w, p) - F(z, p)}{w - z}, \quad w \neq z; \\ G(z, z, p) &= f_p'(z).\end{aligned}$$

We prove

Theorem 6.1: G is continuous on $O \times O \times X$.

PROOF: The Cauchy integral formula for a disk plays an essential part. We consider continuity at a point for which the first two components are equal,

for the continuity is evident in the other case. Let $p, q \in X$, let $\overline{\Delta(a; 2r)} \subset O$, $0 < r < +\infty$, and let $b, c \in \Delta(a; r)$, $b \neq c$. We have

$$\begin{aligned} G(b, c, q) - G(a, a, p) &= \frac{1}{c - b} \left\{ \int_{\vec{bc}} [f_q'(z) - f_p'(a)] dz \right\} \\ &= \frac{1}{c - b} \int_{\vec{bc}} \left[\frac{1}{2\pi} \int_0^{2\pi} \frac{F(z + re^{i\theta}, q) - F(a + re^{i\theta}, p)}{re^{i\theta}} d\theta \right] dz. \end{aligned} \quad (6.1)$$

From (6.1) we conclude that $|G(b, c, q) - G(a, a, p)|$ does not exceed r^{-1} times the value of the modulus of continuity of $F | \overline{\Delta(a; 2r)} \times X$ at

$$\max\{|b - a|, |c - a|\} + \rho(p, q).$$

Reference should be made to Ex. 5.10, Ch. III. We employ the metric in $O \times X$ for which the distance between (z, p) and (w, q) is $|w - z| + \rho(p, q)$. It induces the topology assigned to the product space. Similarly, in $O \times O \times X$ the metric is that which assigns the distance

$$|w_1 - z_1| + |w_2 - z_2| + \rho(p, q)$$

between (z_1, z_2, p) and (w_1, w_2, q) . We have used the fact that if $z \in [b, c]$, then

$$|z - a| \leq \max\{|b - a|, |c - a|\}.$$

We conclude that given $(a, p) \in O \times X$ and η positive there exists δ positive such that when

$$|b - a| + |c - a| + \rho(p, q) < \delta,$$

we have

$$|G(b, c, q) - G(a, a, p)| < \eta$$

subject to the stipulation $b \neq c$. The case $b = c$ now follows by referring to the definition of G and taking the appropriate limit. The theorem follows.

There is an important *corollary*. We note that $G | \overline{\Delta(a; r)} \times \overline{\Delta(a; r)} \times X$ is *uniformly continuous*. Hence, given η positive there exists δ , $0 < \delta < r$, such that

$$|G(a, b, p) - G(a, a, p)| < \eta, \quad p \in X, \quad (6.2)$$

when $|b - a| < \delta$. We see that (6.2) implies the *uniformity with respect to p of the differentiation process*.

Let us apply the results we have obtained to the special case where

$X = [0, 1]$ and consider functions generated from F by integration. Specifically, let $\mu: [0, 1] \rightarrow \mathbb{C}$ have finite variation. Let

$$f(z) = \int_0^1 F(z, t) d\mu(t), \quad z \in O.$$

Thanks to (6.2) we see by direct examination that for each $a \in O$ we have the existence of $f'(a)$ and the equality

$$f'(a) = \int_0^1 G(a, a, t) d\mu(t).$$

It follows that f is *analytic* on O .

EXERCISES

6.1. Let $F_1(z, p) = G(z, z, p)$, $z \in O$, $p \in X$. Then F_1 is continuous on $O \times X$.

6.2. Given f analytic on an open set $O \subset \mathbb{C}$. Suppose that K is a nonempty compact subset of O . Let r be a positive number less than $\text{dist}(K, \mathbb{C} - O)$. We define

$$g(z, w) = [f(z + w) - f(z)]w^{-1}, \quad z \in K, \quad 0 < |w| \leq r,$$

$$g(z, 0) = f'(z), \quad z \in K.$$

Show that g is uniformly continuous. Show more generally that if $n \in \mathbb{N}$, the same holds true for g_n with domain $K \times \overline{\Delta(0; r/n)}$ defined by

$$g_n(z, w) = \left[\sum_{k=0}^n \binom{n}{k} (-1)^{n-k} f(z + kw) \right] w^{-n}, \quad w \neq 0;$$

$$g_n(z, 0) = f^{(n)}(z).$$

6.3. Let $f: [a, b] \rightarrow \mathbb{C}$, $-\infty < a < b < +\infty$. We suppose that $f'(x)$ exists, $a \leq x \leq b$, one-sided derivatives being employed at a and b . Let

$$g(x, y) = \frac{f(y) - f(x)}{y - x}, \quad a \leq x, y \leq b, \quad x \neq y,$$

and let

$$g(x, x) = f'(x), \quad a \leq x \leq b.$$

Show that g is continuous on $[a, b] \times [a, b]$ if and only if f' is continuous on $[a, b]$.

6.4. Let $n \in \mathbb{N}$. Let X be a compact subset of $\mathbb{C}$ and let $O = \mathbb{C} - X$. We introduce $F(z, w) = (w - z)^{-n}$, $(z, w) \in O \times X$. Clearly Theorem 6.1 of the present section is applicable. However it is desired to show directly that, given K compact $\subset O$, the following holds: For each positive η , there exists δ positive such that

$$\left| \frac{F(\zeta, w) - F(z, w)}{\zeta - z} - n(w - z)^{-n-1} \right| < \eta$$

when $(z, w) \in K \times X$, $0 < |\zeta - z| < \delta$. The elementary identity,

$$A^n - B^n = (A - B) \sum_{k=0}^{n-1} A^{n-1-k} B^k,$$

is of service. Unessential computation is to be avoided.

6.5. We consider the expansion of $F(z, p)$ of Theorem 6.1, this section, about $c \in O$ for given $p \in X$, namely

$$F(z, p) = \sum_0^{\infty} A_k(p) (z - c)^k.$$

Let $0 < r < +\infty$ and suppose that $\overline{\Delta(c; r)} \subset O$. Then the series converges uniformly and absolutely on $\Delta(c; r) \times X$. Show that if X is a segment ($\subset \mathbf{R}$) containing a and

$$\lim_a \frac{F(z, t) - F(z, a)}{t - a}$$

exists for each $z \in O$, the limit process being uniform on each compact subset of O , then $A_k'(a)$ exists, $k \in \mathbf{N}$, and

$$F_k(z, a) = \sum_0^{\infty} A_k'(a) (z - c)^k, \quad z \in \Delta(c; r) \subset O.$$

[Bôcher, loc. cit.]

We now turn to some special cases of integrals of the type considered which are of particular importance. The first is *the integral of Cauchy type*. The data consist of a rectifiable path γ in $\mathbf{C}$ and $g: E \rightarrow \hat{\mathbf{C}}$, $\gamma \subset E \subset \hat{\mathbf{C}}$, such that $g \circ \gamma$ takes values in $\mathbf{C}$ and is continuous on its domain.† By the *integral of Cauchy type associated with* (γ, g, z) , $z \in \mathbf{C} - \gamma$, is meant

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} g(w) (w - z)^{-1} dw. \quad (6.3)$$

From the preceding developments of the present section we see that f is analytic on $\mathbf{C} - \gamma$ and by an evident inductive argument that

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_{\gamma} g(w) (w - z)^{-n-1} dw, \quad z \in \mathbf{C} - \gamma, \quad n \in \mathbf{N}.$$

Let us replace γ by a rectifiable closed curve Γ in $\mathbf{C}$ and specialize g to be the constant 1 on $\mathbf{C}$. We see that the function f given by the integral

$$f(z) = \frac{1}{2\pi i} \int_{\Gamma} (w - z)^{-1} dw$$

† We recall the notational license used in the case of paths.

has a derivative equal to zero at each point of $\mathbb{C} - \Gamma$, since for fixed $z \in \mathbb{C} - \Gamma$, $(w - z)^{-2}$ is the derivative of $-(w - z)^{-1}$, $w \in \mathbb{C} - \{z\}$. However, we may establish a much sharper statement:

$$f(z) = O(z; \Gamma), \quad z \in \mathbb{C} - \Gamma.$$

To see this, we fix $a \in \mathbb{C} - \Gamma$ and consider $\Delta(b; r) \subset \mathbb{C} - \{a\}$. If γ is a rectifiable arc with domain $[\alpha, \beta]$ lying in $\Delta(b; r)$ and l is an analytic logarithm of $z - a$, $z \in \Delta(b; r)$, we have

$$\int_{\gamma} (w - a)^{-1} dw = l[\gamma(\beta)] - l[\gamma(\alpha)].$$

To treat our assertion for Γ , we let $m = \min_{\mathbb{R}} |\Gamma - a|$ and take n a positive integer so large that $\omega_{\Gamma}(n^{-1}) < m$. We see that

$$\Gamma(t) \in \Delta[\Gamma(kn^{-1}); m], \quad kn^{-1} \leq t \leq (k+1)n^{-1}.$$

On considering the contribution to

$$\frac{1}{2\pi i} \int_{\Gamma} (w - a)^{-1} dw \tag{6.4}$$

from $[kn^{-1}, (k+1)n^{-1}]$ and summing, we obtain the value

$$\frac{1}{2\pi} \sum_0^{n-1} [\alpha_k(t_{k+1}) - \alpha_k(t_k)],$$

where α_k is a continuous argument of $\Gamma(t) - a$, $kn^{-1} \leq t \leq (k+1)n^{-1}$, $k = 0, \dots, n-1$, the contributions from the real part of the summands being zero because of the periodicity of Γ . Now by Ex. 5.4, this chapter, we see that (6.4) is simply $O(a; \Gamma)$. We now see how the notion of order enters the Cauchy theory significantly (cf. the terminal remarks of §3, this chapter).

EXERCISES

6.6. Let f be analytic on A , a holomorphically simply connected region in $\mathbb{C}$. Let γ denote a closed rectifiable curve in A and let $a \in A - \gamma$. Then

$$O(a; \gamma) f(a) = \frac{1}{2\pi i} \int_{\gamma} f(z) (z - a)^{-1} dz.$$

6.7. Let γ denote a closed rectifiable curve in $\mathbb{C}$. Show that $O(a; \gamma) = 0$ for $|a|$ large proceeding from

$$\int_{\gamma} (w - a)^{-1} dw = -a^{-1} \int_{\gamma} \sum_0^{\infty} \left(\frac{w}{a}\right)^k dw.$$

The analyticity of $O(a; \gamma)$ on $C - \gamma$ implies the vanishing of $O(a; \gamma)$ on the unbounded component of $C - \gamma$.

6.8. Given

$$f(z) = \frac{1}{2\pi i} \int_{\gamma} \frac{g(w)}{w - z} dw, \quad z \in C - \gamma,$$

an integral of Cauchy type. Let $a \in C - \gamma$. We have

$$f(z) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (z - a)^k = \frac{1}{2\pi i} \int_{\gamma} \frac{g(w)}{w - z} \left(\frac{z - a}{w - a} \right)^{n+1} dw.$$

6.9. Calculate for n a positive integer

$$\frac{1}{2\pi i} \oint_{C(0;1)} \frac{w^{-n}}{w - z} dw$$

for $|z| < 1$, for $|z| > 1$.

6.10. Given $g: E \rightarrow C$, $[0, 1] \subset E \subset C$, g continuous on E , g analytic at each point of $]0, 1[$, and γ the identity map of $[0, 1]$ onto itself. Show that the function f defined by the integral of Cauchy type associated with (γ, g, z) has the property that for each $a \in]0, 1[$ there exists φ analytic on a disk centered at a such that $\varphi \cup f_+$ is analytic where $f_+ = f|_{\{\text{Im } z > 0\}}$. A similar result holds for $f_- = f|_{\{\text{Im } z < 0\}}$.

6.11. Calculate

$$\int_{0 \rightarrow 1} (t - z)^{-1} dt, \quad z \in C - [0, 1],$$

and study its limiting behavior at points of $[0, 1]$.

Two other classical instances of integrals involving parameters which occur in the theory of analytic functions are those of Euler and Laplace. They appear in the study of certain linear ordinary differential equations of the second order which arise in problems of mathematical physics. It is not our intention to develop the theory of these integrals in detail but rather just to illustrate the use of integrals representing analytic functions. For details concerning the Euler and Laplace integrals, cf. [34].

The Euler integral. It will be seen that the subject offers excellent experience in the use of continuous and analytic powers. Our data consist of the following: $\alpha \in C$, γ a rectifiable path in C with domain $[0, 1]$, a disk $\Delta(a; r)$ having no points in common with γ , $g: E \rightarrow \hat{C}$, where $\gamma \subset E \subset \hat{C}$, and $g \circ \gamma$ is a continuous map of $[0, 1]$ into C . In what follows we are treating the matter locally when $\alpha \notin Z$. One writes formally

$$\int_{\gamma} g(w) (w - z)^{\alpha} dw,$$

but naturally the notation must be interpreted. What is intended by an *Euler integral associated with* (γ, g, α, z) , $z \in \Delta(a; r)$, is meant

$$\int_0^1 g \circ \gamma(t) K(z, t) d\gamma(t)$$

where K is a continuous α th power of $\gamma(t) - z$, $(z, t) \in \Delta(a; r) \times [0, 1]$. It is not hard to see that such K exist. In fact,

$$K(z, t) = \exp[\alpha L(z, t)]$$

where L is a continuous logarithm of $\gamma(t) - z$, $(z, t) \in \Delta(a; r) \times [0, 1]$. For each $t \in [0, 1]$ the function assigning $L(z, t)$ to $z \in \Delta(a; r)$ is analytic. We conclude that K is analytic in z on $\Delta(a; r)$ for each $t \in [0, 1]$ and further that

$$K_z(z, t) = -\alpha K(z, t) [\gamma(t) - z]^{-1},$$

K_z denoting the derivative of K with respect to z . For the derivative of the function f given by the above Euler integral we have the formula

$$f'(z) = -\alpha \int_0^1 g \circ \gamma(t) \frac{K(z, t)}{\gamma(t) - z} d\gamma(t),$$

which is useful in practice. For specific applications, cf. [34].

The Laplace integral. Here the data consist of γ and g as above. By the *Laplace integral associated with* (γ, g, z) , $z \in \mathbb{C}$, is meant

$$f(z) = \int_{\gamma} g(w) e^{zw} dw. \quad (6.5)$$

Here no questions of interpretation arise. The function f is analytic on $\mathbb{C}$. It is immediate that

$$f'(z) = \int_{\gamma} w g(w) e^{zw} dw.$$

If g is analytic at each point of γ , we have

$$z f(z) = g[\gamma(1)]e^{z\gamma(1)} - g[\gamma(0)]e^{z\gamma(0)} - \int_{\gamma} g'(w) e^{zw} dw.$$

It is to be remarked that we may consider e^{-zw} in place of e^{zw} , as is often done. The two exercises that follow show how the Laplace integral may be applied to the study of certain differential equations. It will be seen that skill in choice of γ and g plays an important part in making the method effective.

EXERCISES

6.12. Let $n \in \mathbb{N}$, $a_k \in \mathbb{C}$, $k = 0, \dots, n+1$, $a_{n+1} \neq 0$. We seek solutions of

$$\sum_0^{n+1} a_k y^{(k)} = 0 \quad (6.6)$$

of the form (6.5) by a suitable choice of γ and g . Let

$$P(z) = \sum_0^{n+1} a_k z^k = a_{n+1} \prod_1^v (z - \alpha_j)^{\mu_j}.$$

Here $\alpha_1, \dots, \alpha_v$ are distinct complex numbers, $\mu_1, \dots, \mu_v$ are positive integers. Clearly f is a solution of (6.6) if and only if

$$\int_{\gamma} P(w) g(w) e^{zw} dw = 0.$$

Show that a nontrivial solution lies at hand when $g(w) = (w - \alpha_j)^{-k}$, $k = 1, \dots, \mu_j$, $j = 1, \dots, v$, and $\gamma(t) = re^{2\pi it}$, $|\alpha_j| < r$, and determine the solution explicitly with the aid of the Cauchy integral formulas for the derivatives of an analytic function.

6.13. Suppose that the coefficient a_k of (6.6) is replaced by $a_k + b_k z$, where $a_k, b_k \in \mathbb{C}$. Suppose that g is analytic on $\{\rho < |z| < +\infty\}$, $\rho \in \mathbb{R}^+$, and that γ is a rectifiable closed curve in $\{\rho < |z| < +\infty\}$. Show that f of (6.5) is a solution of (6.6) so modified if and only if

$$\int_{\gamma} \{[P(w) - Q'(w)] g(w) - Q(w) g'(w)\} e^{zw} dw = 0$$

where $Q(z) = \sum_0^{n+1} b_k z^k$.

The case of the Bessel equation of order 0 is interesting here. For it $P(z) = z$ and $Q(z) = z^2 + 1$. Show that if S is the unique analytic square root of $1 + z^2$ with domain $\mathbb{C} - [-i, i]$ satisfying $\lim_{\infty} z^{-1} S(z) = 1$, then $(2\pi i S)^{-1}$ serves for g . Show that

$$J_0(z) = \frac{1}{2\pi i} \oint_{C(0;r)} \frac{e^{zw}}{S(w)} dw, \quad 1 < r < +\infty,$$

and also that

$$J_0(z) = \frac{1}{\pi} \int_{-1}^1 \frac{e^{itz}}{\sqrt{(1-t^2)}} dt.$$

Show that the n th coefficient in the power series expansion of J_0 is

$$\frac{(i)^n}{\pi(n!)} \int_{-1}^1 \frac{t^n}{\sqrt{(1-t^2)}} dt.$$

7. SOME SUFFICIENT CONDITIONS FOR ANALYTICITY

There is a classical theorem of Morera which is in effect a kind of converse to the Cauchy integral theorem. It states:

If f is a continuous map of an open subset O of $\mathbb{C}$ into $\mathbb{C}$ and for every closed rectifiable curve γ in O we have

$$\int_{\gamma} f dz = 0,$$

then f is analytic on O .

The proof is immediate at the present point. Referring to the proof of Theorem 4.1, this chapter, we see that f possesses a primitive. As the derivative of an analytic function f itself is analytic. The hypothesis of the Morera theorem is excessively stringent. The following is sufficient for the analyticity of f on O : for each $a \in O$, there exists $\Delta(a; r) \subset O$ such that $f|_{\Delta(a; r)}$ satisfies the hypothesis of the Morera theorem. Given f a continuous map of O into $\mathbb{C}$ the analyticity of f is assured when for each $a \in O$ there exists $\Delta(a; r) \subset O$ such that either $f|_{\Delta(a; r)}$ satisfies the conclusion of the simplicial Cauchy-Goursat theorem for $\Delta(a; r)$ -admissible T , or else $f|_{\Delta(a; r)}$ satisfies the conclusion of the theorem of Ex. 2.2, this chapter, for all admitted rectangles, $\Delta(a; r)$ replacing O of that theorem.

EXERCISE

7.1. Prove the last assertion. Apply it to show that if (f_n) is a sequence of analytic functions on O tending uniformly in O to f , then f is analytic on O .

We turn now to a theorem that tells us that under certain circumstances "continuity and an adequate amount of analyticity" imply analyticity. To be specific, let $A < B$, $0 < C$. Let

$$D = \{A < \operatorname{Re} z < B; |\operatorname{Im} z| < C\}.$$

Let f be a continuous map of D into $\mathbb{C}$. Then if f is analytic in $D -]A, B[, f$ is analytic on D .

PROOF: It suffices to make the rectangle test. Suppose that $A < a \leq b < B$ and that $-C < c \leq d < C$. We wish to show that the conclusion of Ex. 2.2, this chapter, holds. This is clear when $0 < c \leq d < C$ and hence by continuity when $0 \leq c \leq d < C$. Similarly the conclusion holds when $-C < c \leq d \leq 0$. If $c < 0 < d$, we consider the rectangles $[a, b; c, 0]$ and $[a, b; 0, d]$ and sum the

integrals of f along their positively sensed boundary paths to obtain the desired result.

EXERCISES

7.2. Give a second proof of the theorem just established by showing if $A < a < b < B$ and $0 < c < C$, then

$$f(z) = (2\pi i)^{-1} \int_{B_1} f(w) (w - z)^{-1} dw,$$

$$0 = (2\pi i)^{-1} \int_{B_2} f(w) (w - z)^{-1} dw,$$

when $a < \operatorname{Re} z < b$, $0 < \operatorname{Im} z < c$. Here B_1 is the positively sensed boundary path for the rectangle $[a, b; 0, c]$ and B_2 is that of $[a, b; -c, 0]$, the rectangle being construed as a strip. Sum to obtain

$$f(z) = (2\pi i)^{-1} \int_B f(w) (w - z)^{-1} dw$$

where z is restricted as above and B is the positively sensed boundary of $[a, b; -c, c]$. Show that the last formula is valid also when $a < \operatorname{Re} z < b$, $-c < \operatorname{Im} z < 0$, and that f is analytic at each point of $]a, b[$. We shall see that the reasoning just employed is very effective in other connections such as Runge's theorem. Does one of these two proofs appeal to fewer prior developments than the other?

7.3. Apply the theorem just proved to show that if f is a continuous map of $\{A < \operatorname{Re} z < B, 0 \leq \operatorname{Im} z < C\}$ into $\mathbb{C}$ which is analytic at each interior point and satisfies $f(x) = 0$, $a < x < b$, then f is the constant 0.

8. RUNGE'S THEOREM

The Cauchy theorem for a rectangle and Cauchy formula for a rectangle taken together serve as a base for a proof of an important approximation theorem due to Runge. It may be stated as follows.

Theorem 8.1: *Let f be analytic on an open subset O of $\mathbb{C}$. Let η be positive, and let K be a compact subset of O . Then there exists a function g of the form*

$$P_\infty(z) + \sum_1^v P_k[(z - a_k)^{-1}], \quad z \in \mathbb{C} - \{a_1, \dots, a_v\}, \quad v \in \mathbb{N},$$

where P_∞ and the P_k are polynomials and the a_k are points of $\operatorname{fr} O$ (fr in the sense of the topology of $\mathbb{C}$), such that

$$|f(z) - g(z)| < \eta, \quad z \in K.$$

PROOF: The case $O = \mathbb{C}$ will be put aside since the power series representation of f yields the desired result. The case where $K = \emptyset$ is vacuously true and will be put aside. We start by introducing r positive such that $K \subset \Delta(0; r)$. Thereupon we introduce

$$m = \text{dist}(K, (C - O) \cup C(0; r)).$$

We take n a positive integer such that $n^{-1}\sqrt{2} < m$. We next introduce the squares

$$S_{jk} = [n^{-1}j, n^{-1}(j+1); n^{-1}k, n^{-1}(k+1)], \quad j, k \in \mathbb{Z},$$

and consider the index pairs (j, k) for which $S_{jk} \subset O \cap \Delta(0; r)$. We let S denote the union of the S_{jk} in question. Clearly $K \subset \text{int } S$, where $\text{int } S$ denotes the set of interior points of S (that is, the interior of S). We let B_{jk} denote the positively sensed boundary path of S_{jk} . From the Cauchy integral theorem and integral formula for a rectangle we have

$$\begin{aligned} \frac{1}{2\pi i} \int_{B_{jk}} f(w) (w - z)^{-1} dw &= f(z), & z \in \text{int } S_{jk}; \\ &= 0, & z \notin S_{jk}. \end{aligned}$$

On summing over the allowed index pairs (j, k) we conclude that

$$f(z) = \sum_1^p \frac{1}{2\pi i} \int_{\gamma_l} f(w) (w - z)^{-1} dw, \quad z \in \bigcup (\text{int } S_{jk}). \quad (8.1)$$

Here the union is taken over allowed (j, k) and the γ_l , $l = 1, \dots, p$, are those sensed edges $\overrightarrow{ab}$ (as in Ex. 2.2, this chapter) of the B_{jk} for which there is not an allowed (λ, μ) with $\overrightarrow{ba}$ a sensed edge of $B_{\lambda\mu}$. The distance between each γ_l and $(C - O) \cup C(0; r)$ is no greater than m . There exist, of course, such γ_l . Since $\bigcup_{1 \leq l \leq p} \gamma_l \subset \text{fr } S$, we see using continuity that (8.1) holds when the condition on z is replaced by the weaker condition, $z \in \text{int } S$, and hence in particular for $z \in K$. It follows from Ex. 1.6, this chapter, that f may be approximated uniformly on K by a sum of the form

$$\sum_1^\mu A_k (w_k - z)^{-1} \quad (8.2)$$

where $A_k \in \mathbb{C}$, $w_k \in \text{fr } S$, $k = 1, \dots, \mu$; that is, given η positive, there exists a sum $g(z)$ of the form (8.2) such that $|f(z) - g(z)| < \eta$, $z \in K$.

Now if $b \in \text{fr } S$, there exists $a \in O \cup C(0; r)$ such that $|b - a| < m$. On the other hand if $z \in K$, $|z - a| \geq m$. Using

$$(z - b)^{-1} = (z - a)^{-1} \left(1 - \frac{b - a}{z - a} \right)^{-1} = \sum_0^\infty \frac{(b - a)^k}{(z - a)^{k+1}}, \quad z \in K,$$

and noting that the series in question converges uniformly on K we infer that f may be approximated uniformly on K by a function of the form indicated in the statement of the theorem where $P_\infty = 0$ but the a_k are allowed to lie on $\text{fr } O \cup C(0; r)$. Now if $|a| = r$, it is obvious that $(z - a)^{-(k+1)}$, $k \in \mathbb{N}$, may be approximated uniformly on K by a polynomial. The theorem of Runge follows by replacing each of the terms of (8.2) by a uniform approximant of it on K of the form $P_k[(z - a_k)^{-1}]$ where

$$a_k \in \text{fr } O \cup C(0; r)$$

and P_k is a polynomial and thereupon replacing each $P_k[(z - a_k)^{-1}]$ for which $|a_k| = r$ by a polynomial approximating it uniformly on K .

The ingenious use of the Cauchy integral theorem and formula for a rectangle in the proof is due to Saks and Zygmund [108].

The theorem of Runge in the present form yields the following theorem in which O and f are as in Theorem 8.1.

Theorem 8.2: *If $\hat{C} - O$ is connected, then f possesses a primitive.*

PROOF: If $a \in C - O$ and γ is a closed rectifiable curve in O , then

$$\int_{\gamma} (z - a)^{-1} dz = 2\pi i O(a; \gamma) = 0$$

since a lies in the unbounded component of $C - \gamma$. Hence

$$\int_{\gamma} g dz = 0$$

for each g of the form given in Theorem 8.1 and hence by Theorem 8.1

$$\int_{\gamma} f dz = 0$$

for f analytic on O , γ being compact. It now follows that f has a primitive.

Later (§4, Ch. XIV) we shall see that, under the assumption of Theorem 8.2, f may be approximated uniformly by a polynomial on each compact subset of O . The result to which we are referring is the Runge Polynomial Approximation Theorem.

EXERCISES

8.1. Give an explicit treatment of the estimates involved in the proof of Theorem 8.1.

8.2. Suppose that O is a nonempty open subset of $\mathbb{C}$, that $\bar{O}$ is its closure in the sense of the topology of $\hat{\mathbb{C}}$, that f is a continuous map of $\bar{O}$ into $\mathbb{C}$, analytic in O . Show by appeal to (8.1) extended to $z \in \text{int } S$ that $\max_{\bar{O}} |f|$ is attained on $\text{fr } O$. One first shows that, for given $z \in \text{int } S$, $|f(z)|^n$ may be estimated suitably in terms of the maximum of $|f|$ on $\text{fr } S$. The use of the Cauchy formula in this connection is due to Landau.

8.3. Let O be an open nonempty subset of $\mathbb{C}$. Let $\mathbf{A}$ denote the set of functions analytic on O . Let $D: \mathbf{A} \rightarrow \mathbf{A}$ satisfy (a) $D(f+g) = D(f) + D(g)$, and (b) $D(fg) = D(f)g + fD(g)$, $f, g \in \mathbf{A}$, and also (c) $(D(f_n))$ tends uniformly in O to $D(f)$ when (f_n) tends uniformly in O to f . Show with the aid of Theorem 8.1 that there exists a unique $g \in \mathbf{A}$ such that $D(f) = f'g$, $f \in \mathbf{A}$. The maps taking f into $f'g$ for some $g \in \mathbf{A}$, of course fulfill the stated conditions. D fulfilling (a) and (b) is called a *derivation* of $\mathbf{A}$. Condition (c) is a continuity condition.

9. MAPPING PROPERTIES OF ANALYTIC FUNCTIONS

We turn to the study of the nature of the mapping defined by an analytic function and shall be concerned with three central questions. They are: the action of a nonconstant analytic function on an open set, the local behavior of an analytic function, and the nature of the inverse of a univalent analytic function. Basic instruments of investigation will be the order of a point with respect to a closed curve, its representation as an integral of Cauchy type, the formula for the action of an analytic function on a line integral (Ex. 3.21, this chapter). Useful integral formulas will emerge.

Given f analytic at a , by the *multiplicity of f at a* , denoted $n(a; f)$, is meant

$$\inf \{k: k-1 \in \mathbf{N} \text{ and } f^{(k)}(a) \neq 0\}.$$

Given f analytic on a region A , by the *valence of f at $w \in \hat{\mathbb{C}}$* , denoted $v_f(w)$, is meant $+\infty$ when the set $\{f(z) = w\}$ is infinite, and otherwise the nonnegative integer

$$\sum_{f(z)=w} n(z; f),$$

the sum being zero, of course, when f does not take the value w . We have deliberately made $\hat{\mathbb{C}}$ the domain of v_f by way of anticipation of the situation for meromorphic functions.

Now suppose that f is analytic on a region A and is not constant and that $\Delta(a; r) \subset A$, $0 < r < +\infty$. We let $\gamma(t) = a + re^{2\pi it}$, $t \in \mathbf{R}$, and investigate $O(w; f \circ \gamma)$ for $w \notin f[C(a; r)]$. If w also satisfies $w \notin [\Delta(a; r)]$, then $(f-w)|_{\Delta(a; r)}$ has a continuous logarithm as we see on considering $(f-w)|_{\Delta(a; \rho)}$ where $\rho > r$ and is sufficiently close to r . Since $f \circ \gamma - w$ has a periodic continuous logarithm with period 1, we see that $O(w; f \circ \gamma) = 0$ —a conclusion which can also be arrived at by considering $O(w; f \circ \gamma_t)$, $0 \leq t \leq 1$,

where $\gamma_\tau = (1 - \tau)\gamma + \tau a$. Suppose now that

$$w \in f[\Delta(a; r)] - f[C(a; r)].$$

Let $b_1, \dots, b_m$ denote the points of $\Delta(a; r)$ at which f takes the value w , $b_k \neq b_j$ for $k \neq j$. We have

$$f(z) - w = \left[\prod_1^m (z - b_k)^{n(b_k; f)} \right] g(z), \quad (9.1)$$

where g is analytic on A and takes the value zero nowhere on $\overline{\Delta(a; r)}$ as we see on examining the quotient

$$\frac{f(z) - w}{\prod_1^m (z - b_k)^{n(b_k; f)}},$$

noting that for each $j, j = 1, \dots, m$, the factor $(z - b_j)^{n(b_j; f)}$ may be canceled from numerator and denominator. What remains defines a function analytic at b_j . From (9.1) we conclude

$$O(w; f \circ \gamma) = \sum_1^m n(b_k; f) O(b_k; \gamma) = \sum_1^m n(b_k; f).$$

We are led to the important conclusion: *For each $w \in \mathbf{C} - f[C(a; r)]$, the valence at w of $f| \Delta(a; r)$ is equal to $O(w; f \circ \gamma)$. Hence the valence of $f| \Delta(a; r)$ is constant on each component of $\hat{\mathbf{C}} - f[C(a; r)]$.*

We note that if $f(a) \notin f[C(a; r)]$, then the valence of $f| \Delta(a; r)$ takes a constant positive integer value on the component of $\mathbf{C} - f[C(a; r)]$ which contains $f(a)$ and hence $f[\Delta(a; r)]$ is a neighborhood of $f(a)$. The supposition $f(a) \notin f[C(a; r)]$ is certainly fulfilled when r is small and positive. Consequently, we have the following fundamental conclusion: *If O is an open subset of A , then $f(O)$ is open.* The property asserted is known as the *open mapping property* of nonconstant analytic functions.

EXERCISE

9.1. Show that if p is a polynomial function of positive degree, then $p(\mathbf{C})$ is both an open and closed subset of $\mathbf{C}$, and conclude the fundamental theorem of algebra.

We see from earlier developments (Ex. 5.8, Ch. II) that if $f'(a) \neq 0$, then f is univalent in some neighborhood of a . We shall now give a second proof based on the notion of the order of a point with respect to a closed curve.

Indeed, let us take $r, 0 < r < +\infty$, such that $\overline{\Delta(a; r)} \subset A$ and f attains the value $f(a)$ on $\overline{\Delta(a; r)}$ only at a . We take $s, 0 < s < r$, so small that

$$f[\Delta(a; s)] \subset \Delta[f(a); m]$$

where

$$m = \min\{|f(a + re^{i\theta}) - f(a)| : \theta \in \mathbf{R}\}.$$

Since the valence at w of $f|_{\Delta(a; r)}$ is one, $w \in \Delta[f(a); m]$, we have $f|_{\Delta(a; s)}$ univalent.

In the opposite direction, we are led to ask what can be said about the derivative of a univalent analytic function. It is to be kept in mind that in real analysis the univalence of a function possessing a derivative need not force the derivative to take nonzero values. Thus $f(x) = x^3$, $x \in \mathbf{R}$, is an increasing map of $\mathbf{R}$ onto itself possessing a derivative at each point of its domain, and yet $f'(0) = 0$. Our experience with complex analysis up to this point indicates that as far as differential analysis is concerned we are operating in a domain where the phenomena are simpler than those encountered in the real domain. For this reason we might feel cautiously optimistic about conjecturing that if f is a univalent analytic function, then f' does not take the value 0.

To see that this is so, we consider f analytic on A and nonconstant and take $\overline{\Delta(a; r)} \subset A$, $0 < r < +\infty$, where r is so small that not only does f take the value $f(a)$ on $\overline{\Delta(a; r)}$ only at a but also $f'(z) \neq 0$, $0 < |z - a| < r$. It follows that every point of the component of $\mathbf{C} - f[C(a; r)]$ which contains $f(a)$, save $f(a)$ itself, is attained at $n(a; f)$ distinct points of $\overline{\Delta(a; r)}$. Hence, if f is univalent, then $n(a; f) = 1$, that is, $f'(a) \neq 0$.

The exponential function shows that the converse is false.

An integral representation of the order of a point with respect to $f \circ \gamma$. Here f and γ are understood in the sense of the third paragraph of this section. We know from §6, this chapter, that

$$O(b; f \circ \gamma) = \frac{1}{2\pi i} \int_{f \circ \gamma} (w - b)^{-1} dw, \quad b \in \mathbf{C} - f[C(a; r)].$$

Applying the result of Ex. 3.21, this chapter, with f of the exercise taken over by $(w - b)^{-1}$ and g of that exercise by f restricted to $f^{-1}(\mathbf{C} - \{b\})$, we find that

$$O(b; f \circ \gamma) = \frac{1}{2\pi i} \oint_{C(a; r)} \frac{f'(z)}{f(z) - b} dz. \quad (9.2)$$

We pass from (9.2) to a cognate integral that is veritably a *deus ex machina* in the study of the inverse of an analytic function. We introduce

$$g(w) = \frac{1}{2\pi i} \oint_{C(a; r)} \frac{z f'(z)}{f(z) - w} dz, \quad w \in \mathbb{C} - f[C(a; r)]. \quad (9.3)$$

We know from the results of §6, this chapter, that the function g is analytic on its domain and by the Cauchy integral theorem for a disk that $g(w) = 0$ when $w \notin f[\Delta(a; r)]$.

We now suppose that $f'(a) \neq 0$ and that $f(z) \neq f(a)$, $0 < |z - a| \leq r$. Let B denote the component of $\mathbb{C} - f[C(a; r)]$ which contains $f(a)$. Given $b \in B$, let c denote the unique point of $\Delta(a; r)$ such that $f(c) = b$. Then

$$f(z) - b = \varphi(z)(z - c),$$

where φ is analytic on A , $\varphi(c) = f'(c)$, and $\varphi(z) \neq 0$, $z \in \overline{\Delta(a; r)}$. Hence

$$g(b) = \frac{1}{2\pi i} \oint_{C(a; r)} \frac{z f'(z)}{\varphi(z)} (z - c)^{-1} dz = c.$$

We conclude that

$$f[g(w)] = w, \quad w \in B.$$

Suppose now that, in addition, $f|_{\overline{\Delta(a; r)}}$ is univalent. Then $f[\Delta(a; r)] = B$. For $f[\Delta(a; r)]$ is connected, contains $f(a)$, and lies in $\mathbb{C} - f[C(a; r)]$. Hence $f[\Delta(a; r)] \subset B$ and, since $B \subset f[\Delta(a; r)]$, equality holds. From the equality we see that $g|_B$ is the inverse of $f|_{\Delta(a; r)}$. We are led to

Theorem 9.1: *Let f be analytic on O , an open subset of $\mathbb{C}$, and be univalent. Then the inverse of f is analytic on its domain.*

The only comment to be made is that the above local argument applies to $f|_{\Delta(a; r)}$ where $\overline{\Delta(a; r)} \subset O$, $0 < r < +\infty$, and that $g|_B = f^{-1}|_B$. Hence f^{-1} is analytic.

EXERCISES

9.2. Reference is made to (9.3). For $b \in \mathbb{C} - f[C(a; r)]$ we have

$$g^{(k)}(b) = \frac{k!}{2\pi i} \oint_{C(a; r)} \frac{z f'(z)}{[f(z) - b]^{k+1}} dz, \quad k \in \mathbb{N}.$$

9.3. Continuing, suppose that $f'(a) \neq 0$, that $f(z) \neq f(a)$, $0 < |z - a| \leq r$, and that,

with $b = f(a)$, the series

$$\sum_0^{\infty} \frac{g^{(k)}(b)}{k!} (z-b)^k$$

converges on $\Delta(b; \rho)$ and has sum G on $\Delta(b; \rho)$, and that $G[\Delta(b; \rho)] \subset A$, the domain of f . Then $f[G(z)] = z$, $|z-b| < \rho$.

9.4. We suppose f and h analytic at each point of $\overline{\Delta(a; r)}$, $0 < r < +\infty$, $f'(a) \neq 0$, $f(z) \neq f(a)$, $0 < |z-a| \leq r$. Evaluate

$$\frac{1}{2\pi i} \oint_{C(a; r)} h(z) \frac{f'(z)}{f(z) - w} dz,$$

where $w \in B$ (as in text). What is the value under the hypothesis $w \in C - f[C(a; r)]$?

9.5. Suppose that φ is analytic on a region A , $0 \in A$, and that $\varphi(z) \neq 0$, $z \in A$. Let $f(z) = z \varphi(z)$, $z \in A$. Calculate $g^{(k)}(0)$, $k \in \mathbb{N}$, for g associated with small positive r in terms of φ . Determine the $g^{(k)}(0)$ for the following cases:

$$\varphi(z) = e^z;$$

$$\varphi(z) = (e^z - 1)z^{-1}, \quad z \neq 0, \quad \varphi(0) = 1;$$

$$\varphi(z) = 1 - \varepsilon(\sin z)z^{-1}, \quad z \neq 0, \quad \varphi(0) = 1 - \varepsilon, \quad 0 < \varepsilon < 1.$$

The last example arises from Kepler's equation.

9.6. Let f be analytic on a region A and be univalent. Using the open mapping property of f and the fact that $f'(z) \neq 0$, $z \in A$, show that g , the inverse of f , is analytic, the argument being based on the continuity of g and the equality $f(z) - b = \varphi(z)(z-a)$, $b = f(a)$, φ continuous on A and $\varphi(z) \neq 0$, $z \in A$ (cf. Theorem 1.1 of Ch. IV).

9.7. Show that we may even dispense with the information that $f'(z) \neq 0$, $z \in A$, in proving the analyticity of g . It is to be noted that at all events $E = \{z: f'(z) = 0\}$ clusters at no point of A and hence its image under f clusters at no point of $f(A)$ by virtue of the univalence and open mapping property of f . The analyticity of g at the points of $f(A - E)$ is to be cared for first. How do we infer the analyticity of g on $f(A)$ and the fact that $E = \emptyset$?

9.8. Show that the maximum principle of §8, Ch. IV, holds when we consider the modulus of an analytic function in place of a harmonic function. Conclude that if the modulus of a function f analytic on a region attains its maximum, then f is constant (cf. Exs. 7.2 and 7.5, Ch. IV.) Obtain the open mapping property for analytic functions with its aid as follows. Starting with $f(z) \neq f(a)$, $z \in C(a; r)$, let

$$m = \min \{|f(a + re^{i\theta}) - f(a)|: \theta \in \mathbb{R}\}$$

and consider $P_b(z) = |f(z) - b|$, $z \in \overline{\Delta(a; r)}$, $|b - f(a)| < m/2$. Show that

$$P_b(a) < \min \{|f(a + re^{i\theta}) - b|: \theta \in \mathbb{R}\}$$

and examine z minimizing P_b .

9.9. Let $n \in \mathbb{N}$. Let $b_0, \dots, b_n$ be analytic on $\Delta(a; \rho)$. Let

$$P(w, z) = w^{n+1} + \sum_0^n b_k(z)w^k.$$

Suppose that $P(c, a) = 0$ and that $P_w(c, a) \neq 0$. [The subscript w indicates the partial derivative of P with respect to the first variable, the definition of the notion of partial derivative being a direct paraphrase of that given for functions of two real variables.] Now let r be positive and so small that $P(w, a) \neq 0$ for $0 < |w - c| \leq r$ and let s be positive and so small that $P(w, z) \neq 0$ for $|w - c| = r$ and $|z - a| < s$. Show that

$$g = \{(z, w): (z, w) \in \Delta(a; s) \times \Delta(c; r), P(w, z) = 0\}$$

is an analytic function with domain $\Delta(a; s)$. Use may be made of the integrals

$$\frac{1}{2\pi i} \oint_{C(c; r)} w^k \frac{P_w(w, z)}{P(w, z)} dw, \quad k = 0, 1.$$

Note that they define functions analytic in $\Delta(a; s)$.

***9.10.** Let f be analytic on an open subset O of $\mathbb{C}$, $f(z) \neq 0$, $z \in O$. Let g be analytic on O . Then a continuous g th power of f is analytic.

Local behavior of a nonconstant analytic function. As the final matter to be treated in this chapter we consider the question of the local behavior of an analytic function. Let f be analytic at a with $n(a; f) < +\infty$. We prove

Theorem 9.2: *There exists positive r such that $\Delta(a; r)$ is contained in the domain of f and*

$$f(z) = f(a) + [g(z)]^{n(a; f)}, \quad z \in \Delta(a; r),$$

where g is analytic on $\Delta(a; r)$ and univalent.

PROOF: We take ρ positive so that f is analytic in $\Delta(a; \rho)$ and

$$f(z) - f(a) = \varphi(z) (z - a)^{n(a; f)}, \quad z \in \Delta(a; \rho),$$

where φ is analytic on $\Delta(a; \rho)$ and $\varphi(z) \neq 0$, $z \in \Delta(a; \rho)$. Let ψ denote an analytic $n(a; f)$ th root of φ . We take $g(z) = \psi(z) (z - a)$ with domain $\Delta(a; r)$, $0 < r \leq \rho$, where r is so chosen that g is univalent. The theorem follows.

We shall see that this theorem has many important consequences in the study of the maps given by analytic (and meromorphic) functions. On referring to the properties of the map z^v , $z \in \mathbb{C}$, v a positive integer, we obtain the following corollary of Theorem 9.2.

Corollary: *With r so chosen, $n(a; f)$ is the maximum of the valence of $f| \Delta(a; r)$. It is attained on a neighborhood of b .*

In fact, it is immediate that $n(0; z^v) = v$, $n(b; z^v) = 1$, $b \in \mathbb{C} - \{0\}$. Now $f| \Delta(a; r)$ takes the value $f(a)$ exactly at a . The valence at $f(a)$ is $n(a; f)$. If $w \neq f(a)$, then $f(z) = w$, $z \in \Delta(a; r)$, if and only if $g(z)$ is an $n(a; f)$ th root of

$w = f(a)$. The multiplicity of f at each such z is 1. Hence the valence in question is at most $n(a; f)$. Since g is an open map, its image contains a disk $\Delta(0; \epsilon)$ and we conclude that the valence in question is $n(a; f)$ at each point of $\Delta[f(a); \epsilon^{n(a; f)}]$.

EXERCISES

9.11. Give a power-series-theoretic proof of Theorem 9.2, using

$$f(z) - f(a) = (z - a)^{n(a; f)} \varphi(a) \left[1 + \frac{\varphi(z) - \varphi(a)}{\varphi(a)} \right],$$

z near a , the binomial series, the composition theorem for power series, and the local univalence near 0 of the sum of a power series with positive radius of convergence whose derivative at 0 is not zero.

***9.12.** Let f be analytic at each point of $\overline{\Delta(0; 1)}$. Suppose that $\operatorname{Re} f(e^{2\pi i t})$ has $2k$ changes of sign on $[0, 1[$. Then the valence at 0 of $f|_{\Delta(0; 1)}$ is at most k (Backlund). Hint: Cf. Ex. 5.8, this chapter. Make circular detours about points on $C(0; 1)$ at which f takes the value 0.